The birth–death–catastrophe process: distribution theory, quasi-stationarity, and burst statistics

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1 Introduction

The standard models of within-host infection treat the infected cell as a black box. In the Basic Model of Viral Replication, a cell that has been infected releases new particles at a constant rate p until it is removed at a constant rate $d_{\mathcal{I}}$; nothing about the cell's interior enters the dynamics, and the two constants are phenomenological — parameters to be fitted, not quantities to be derived. For a budding virus this is at least a defensible caricature. For a lytic pathogen it is plainly wrong in mechanism, whatever it may achieve by fitting: such a pathogen grows inside the cell, and releases nothing until the cell ruptures, at which point it releases everything at once. *Yersinia pestis* in the macrophage, *Francisella tularensis* in the macrophage, the lytic phage in its bacterium — all follow the intracellular replication-and-lysis picture, and all are poorly served by a constant drip.

Any replacement of that black box must begin with a complete description of a single infected cell, and that description is the subject of this chapter. The intracellular model was set up in Chapter 3: the birth–death–catastrophe process, in which the pathogen count within one cell is born at rate λ per particle, dies at rate μ per particle, and triggers the rupture of the cell at rate δ per particle. Chapter 3 derived the principal single-cell quantities — the no-rupture probability $I(t)$, the non-fixation probability $I_{\text{fix}}(t)$, the mean and second moment $J(t)$, $K(t)$, and the cumulative mean release $V(t)$ — each in closed form. What it did not derive is the full distribution of the process, or the statistics of the rupture event itself. These are the subjects of the present chapter.

Three questions organise it.

1. What is the complete single-cell description? The state probabilities at every time, the quasi-stationary distribution of the intracellular load, and the laws of the burst time and the burst size (§§3–8).

2. How do several founders in one cell behave? The multi-founder quantities foreshadowed in Chapter 3, and the conditional means available to an experiment that observes a burst (§8–§9).

3. What happens when the released particles found a new infection immediately? The chained immediate-transfer model, in which one cell's rupture is the next cell's inoculum (§10).

The answers, in advance and in brief, are these. The single-cell process is solvable far beyond its moments: the state probabilities are geometric at every time, and their conditional limit — the quasi-stationary distribution — is geometric with ratio $1/a$. The burst-size distribution, conditioned on bursting, is *exactly the same law*: a cell's rupture size is distributed as the load it would have had, had it kept growing forever. Burst times given burst sizes obey a harmonic-number law; cells infected by k particles release superlinearly more, and the chained transfer of bursts from cell to cell has an exact negative-binomial structure. Each of these results is stated in Chapter 3's notation, which this chapter completes rather than revises.

The chapter that follows this one, Chapter 4b, takes the completed single-cell theory and embeds it in a population: the survival and release kernels derived here become the age-structured kernels of a renewal model of viral dynamics, and the constant-rate black box of the Basic Model is shown to be a projection of that model, with quantifiable distortions. Nothing beyond the contents of the present chapter is needed for that step.

1.1 Two varieties of rupture: killing and catastrophe

Before we begin, it is important to state a distinction between two varieties of the model, already touched on in Chapter 3. When a rupture event occurs, the process goes to a state. In the literature, this state is sometimes chosen to be 0, and in other cases there is a separate rupture state defined, sometimes labelled H. The analysis does differ between the two options. And so for clarity, in this chapter we will refer to the 0-state case as “killing”; the distinct rupture state as “catastrophe”. The distinction matters most in §§3–5, where both conventions are carried in parallel; from §6 onward the catastrophe convention is primary.

2 What we need from Chapter 3

Chapter 3 defined the birth–death–catastrophe process and derived its principal single-cell quantities in closed form. This chapter builds on that foundation, and everything it needs is restated here, boxed, so that what follows is self-contained. Derivations are at most sketched; the full accounts live in Chapter 3, to which each box points. Section 2.1 fixes the notation used throughout; the remaining subsections collect the closed forms that will be quoted repeatedly.

2.1 Canonical notation

The notation of this chapter is collected in Table 1; it is the notation of Chapter 3, unchanged. When, in Chapter 4b, the single-cell theory is joined to a population model, further symbols are introduced there, chosen to avoid collisions with these.

Symbol	Meaning	Notes
λ, μ, δ	birth, death, catastrophe rates (per capita)	
$\mathbf{X}_t$	intracellular count (CTMC)	
H	absorbing rupture state	
$\mathbf{W}_t$	released count from one cell	0 until burst
τ	burst time (possibly ∞)	
$\mathcal{K}$	burst size (random variable)	never bare K, since $K(t) = \mathbb{E}({\mathbf{X}_t}^2)$
$I(t)$	$\mathbb{P}\{\text{no burst by } t\}$	
$D(t)$	$\mathbb{P}\{\text{cleared internally, no burst}\}$	$= p_0(t)$
$I_{\text{fix}}(t)$	$\mathbb{P}\{\mathbf{X}_t \notin \{0, H\}\} = I(t) - D(t)$	productive infection
$J(t), K(t)$	$\mathbb{E}(\mathbf{X}_t)$, $\mathbb{E}({\mathbf{X}_t}^2)$	
$V(t)$	$\mathbb{E}(\mathbf{W}_t)$, cumulative mean release	
a, b	characteristic roots, $b < 1 < a$	
A, B, θ	$a - 1$, $1 - b$, $\lambda(a - b)$	
κ	$1 + \delta/(2\lambda)$	appears in $K(t)$ and $K(I)$
w	$\mathrm{e}^{\theta t}$	the recurring exponential

Table 1: Canonical notation for this chapter (Chapter 3’s regime).

2.2 Process definition and the joint process

One cell, one intracellular population. The count $\mathbf{X}_t$ is a continuous-time Markov chain on $\{0, 1, 2, \ldots\} \cup \{H\}$ which, from any state $n \geq 1$, jumps

$$n \to n + 1 \qquad \text{at rate } \lambda n \quad \text{(birth)},$$

$$
\begin{aligned}
n &\to n-1 &&\text{at rate } \mu n \quad \text{(death)},\\
n &\to H &&\text{at rate } \delta n \quad \text{(catastrophe)}.
\end{aligned}
$$

Both 0 and H are absorbing: the population either clears itself internally, or the cell ruptures. Unless stated otherwise we start from a single founder, $X_0 = 1$. The process of classical references [1, 2, 3] is slightly more general (immigration, distributed catastrophe sizes); the specialisation above is the one with the closed-form family we need.

FIGURE FLAG F4a.1 — Joint-process schematic
Purpose: display the object of this chapter — the joint process (X_t, W_t) — at a glance: the stochastic growth of the load, the burst at τ, and the jump of the released count from 0 to X_{τ^-}. **Type:** TikZ schematic, inline with chapter fonts. **Content:** a horizontal time axis; one representative sample path of X_t drawn as a step function starting at $X_0 = 1$, wandering up and down through states $1, 2, \ldots$, terminating in a vertical jump to a marked state H at time τ; below it, on a second axis, the corresponding path of W_t: zero until τ, then a vertical jump to the level X_{τ^-} and a flat continuation thereafter; annotate τ, X_{τ^-}, and label the state H with an ellipse as in Chapter 3's transition diagram. **Labels:** t on the horizontal axis; X_t and W_t on the two vertical axes; annotations τ, X_{τ^-}, H. **Style:** TikZ, inline with chapter fonts, black paths, the burst jump optionally emphasised in colour. **Placement:** Section 2, in the "Process definition and the joint process" subsection, after the transition-rate display.

The rupture event is what matters for applications, so we record the release count alongside the load. Let τ be the time of catastrophe (possibly $\tau = \infty$ if the population clears internally first). The released count $\mathbf{W}_t$ is

$$
\mathbf{W}_t = \begin{cases} 0, & t < \tau, \\ X_{\tau^-}, & t \geq \tau, \end{cases}
$$

that is, at rupture the whole pre-burst load is dumped and then held. All single-cell release quantities in this chapter are functionals of the joint process $(\mathbf{X}_t, \mathbf{W}_t)$.

2.3 Roots and the workhorse identity

Write

$$
\eta = \frac{\lambda + \mu + \delta}{2\lambda}, \qquad a = \eta + \sqrt{\eta^2 - \tfrac{\mu}{\lambda}}, \qquad b = \eta - \sqrt{\eta^2 - \tfrac{\mu}{\lambda}}, \tag{1}
$$

the two roots of the quadratic $\lambda f^2 - (\lambda + \mu + \delta) f + \mu$, and set

$$
A := a - 1 > 0, \qquad B := 1 - b > 0, \qquad \theta := \lambda(a - b) = \sqrt{(\lambda + \mu + \delta)^2 - 4\lambda\mu}. \tag{2}
$$

Vieta's relations give $ab = \mu/\lambda$ and $a + b = (\lambda + \mu + \delta)/\lambda$, and the identity that will do most of the algebraic work in this chapter is

$$
AB = (a-1)(1-b) = (a+b) - ab - 1 = \frac{\delta}{\lambda}, \qquad A + B = a - b. \tag{3}
$$

That $b < 1 < a$ always is immediate from $\lambda(1-a)(1-b) = \lambda\,(1 - (a+b) + ab) = -\delta < 0$, which places 1 strictly between the two roots.

2.4 Three fixation functions

There are three absorbing-behaviour probabilities, not two, and it pays to name all of them:

Definition 2.1.

$$I(t) = \mathbb{P}\{\text{no burst by } t\}\,, \qquad D(t) = \mathbb{P}\{\text{internal extinction by } t,\ \text{no burst}\}\,, \tag{4}$$

$$I_{\text{fix}}(t) = \mathbb{P}\{\mathbf{X}_t \notin \{0, H\}\} = I(t) - D(t), \tag{5}$$

the last being the probability that the cell is still productively infected.

Both I and D satisfy the same Riccati equation

$$\frac{\mathrm{d}I}{\mathrm{d}t} = \lambda(I-a)(I-b) = \mu - (\lambda+\mu+\delta)I + \lambda I^2, \tag{6}$$

differently initialised ($I(0) = 1$, $D(0) = 0$); the branching term I^2 comes from the two independent daughter lineages that a birth of the founder creates. Solving by separation of variables and writing $w := \mathrm{e}^{\theta t} \geq 1$ gives the boxed triplet

$$\boxed{I(t) = \frac{aB + bAw}{B + Aw}, \qquad D(t) = \frac{ab\,(w-1)}{aw - b}, \qquad I_{\text{fix}}(t) = \frac{(a-b)^2\,w}{(B+Aw)(aw-b)}.} \tag{7}$$

The limits are $I(\infty) = D(\infty) = b$, $I_{\text{fix}}(\infty) = 0$, with $I_{\text{fix}}(0) = 1$ and $I'_{\text{fix}}(0) = -(\mu+\delta)$ — at the instant of founding, the productive state is lost only by death or catastrophe of the single founder. An independent confirmation of the third formula arrives in §5: the state probabilities $p_n(t)$ sum over $n \geq 1$ to exactly the right-hand side of (7).

2.5 Moments and mean release

The hazard of bursting at time t is δ times the current load, so $I'(t) = -\delta J(t)$; inserting the Riccati equation (6) then gives the mean

$$\boxed{J(t) = \mathbb{E}(\mathbf{X}_t) = -\frac{1}{\delta}\frac{\mathrm{d}I}{\mathrm{d}t} = \frac{(a-b)^2\,w}{(B+Aw)^2} = -\frac{\lambda}{\delta}(I-a)(I-b),} \tag{8}$$

with $J(0) = 1$ now manifest. The second moment,

$$\boxed{K(t) = \mathbb{E}({\mathbf{X}_t}^2) = \Big[1 + \tfrac{2\lambda}{\delta}(1-I)\Big]J = \frac{2\lambda}{\delta}(\kappa - I)J, \qquad \kappa = 1 + \frac{\delta}{2\lambda},} \tag{9}$$

follows from the moment hierarchy: one checks by differentiation, using (6), that the right-hand side satisfies $K' = 2(\lambda-\mu)K + (\lambda+\mu)J - \delta\mathbb{E}({\mathbf{X}_t}^3)$ with the third moment eliminated exactly as in Chapter 3.

The cumulative mean release $V(t) = \mathbb{E}(\mathbf{W}_t)$ increases only at rupture events, and a rupture from state n occurs at rate δn and releases n particles, so the expected release flux is $\delta\mathbb{E}({\mathbf{X}_t}^2)$:

$$V'(t) = \delta K(t), \qquad V(0) = 0. \tag{10}$$

Integrating with (8)–(9) yields the simplification

$$\boxed{V(t) = (1-I)\Big[1 + \tfrac{\lambda}{\delta}(1-I)\Big],} \tag{11}$$

a quadratic in $(1-I)$ alone — no J, no separate μ. The equivalent Chapter 3 form $V = \frac{\lambda-\mu}{\delta}(1-I) - J + 1$ reduces to this one upon substituting $J = -(\lambda/\delta)(I-a)(I-b)$ and using (3).

The lifetime mean release and the conditional mean burst size are

$$\boxed{V_\infty = \frac{a(1-b)}{a-1} = \frac{aB}{A} = \frac{\lambda-\mu}{\delta}(1-b) + 1, \qquad \mathbb{E}(\mathcal{K} \mid \text{burst}) = \frac{V_\infty}{1-b} = \frac{a}{a-1}.} \tag{12}$$

A small point with large consequences: V_∞ is $\mathbb{E}(\mathcal{K}\mathbf{1}\{\text{burst}\})$, and so *includes* the realisations that cleared internally and released nothing (a mass b at 0). The conditional mean removes them. When $\mu = 0$: $b = 0$, $a = 1 + \delta/\lambda$, and $V_\infty = 1 + \lambda/\delta$ coincides with the conditional mean — a coincidence that, as we shall see, one should not lean on.

2.6 Second moment of the release

For the assay predictions of the discussion we will need $\mathbb{E}({\mathbf{W}_t}^2)$. The release variable jumps once, from 0 to X_{τ^-}, so $\dfrac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}({\mathbf{W}_t}^2) = \delta\mathbb{E}({\mathbf{X}_t}^3)$. The third moment is eliminated via the second-moment ODE, $K' = 2(\lambda-\mu)K + (\lambda+\mu)J - \delta\mathbb{E}({\mathbf{X}_t}^3)$, giving

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}({\mathbf{W}_t}^2) = 2(\lambda-\mu)K + (\lambda+\mu)J - K' = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{2(\lambda-\mu)}{\delta}V - \frac{\lambda+\mu}{\delta}I - K\right),$$

In the second equality we used $V' = \delta K$ together with $J = -\delta^{-1}I'$. Integrating from 0 with $(I, J, K, V, \mathbb{E}(W^2))(0) = (1, 1, 1, 0, 0)$ — in particular $K(0) = \mathbb{E}(X_0^2) = 1$ —

$$\boxed{\mathbb{E}({\mathbf{W}_t}^2) = \frac{2(\lambda-\mu)}{\delta}V - \frac{\lambda+\mu}{\delta}I - K + \frac{\lambda+\mu}{\delta} + 1,} \tag{13}$$

and $\mathrm{Var}(\mathbf{W}_t) = \mathbb{E}({\mathbf{W}_t}^2) - V^2$. As a check, take $\mu = 0$ and $t \to \infty$: then $I, K \to 0$ and $V \to 1 + \lambda/\delta$, so (13) gives $\mathbb{E}(W_\infty^2) = 2\lambda^2/\delta^2 + 3\lambda/\delta + 1$, which is exactly the second moment of the geometric burst-size law of §7.2 with success probability $s = \delta/(\lambda+\delta)$, whose second moment is $(2-s)/s^2$.

3 Probability Generating Function PDEs

The probability generating function (probability generating function) of the intracellular count is defined by

$$G(z,t) = p_0(t) + p_1(t)z + p_2(t)z^2 + p_3(t)z^3 + \ldots,$$

where $p_i(t) = \Pr\{\mathbf{X}_t = i\}$. The following PDEs are derived from the forward Kolmogorov equations associated with the two rupture conventions.

Catastrophe

$$\frac{\partial G(z,t)}{\partial t} = \left(\lambda z^2 - (\lambda+\mu+\delta)z + \mu\right)\frac{\partial G}{\partial z} \tag{14}$$

Killing

$$\frac{\partial G(z,t)}{\partial t} = \delta J(t) + \left(\lambda z^2 - (\lambda+\mu+\delta)z + \mu\right)\frac{\partial G}{\partial z} \tag{15}$$

A word of warning before we proceed. In the catastrophe convention, ruptured realisations are absorbed into the separate state H and no longer contribute to the count, so $G(1,t) = I(t) \le 1$: the probability generating function is *defective*, and its total mass decays as cells burst. In the killing convention the rupture transition lands in state 0 and is counted, so the probability generating function remains proper: $G(1,t) = 1$ for all t. We will keep track of which is which.

Derivation

It will be sufficient to show a derivation of the "killing" equation; the "catastrophe" version is similar and simpler.

Take the forward Kolmogorov equations:

$$\begin{cases} \dfrac{\mathrm{d}p_i}{\mathrm{d}t} = \lambda(i-1)p_{i-1} + \mu(i+1)p_{i+1} - (\lambda+\mu+\delta)ip_i, \quad i > 0 \\[2ex] \dfrac{\mathrm{d}p_0}{\mathrm{d}t} = \mu p_1 + \delta \displaystyle\sum_{j=1}^{\infty} jp_j. \end{cases}$$

($p_i = \Pr\{\mathbf{X}_t = i\}$, the probability the process is in state i). In the latter equation, the summation represents all of the possible rupture transitions: from any positive state, a collapse to 0 can take place. This summation happens to express the mean, $\mathbb{E}(\mathbf{X}_t)$, which is elsewhere labelled $J(t)$.

In the catastrophe version, since rupture causes transition to a separate state, this summation term is not present: the first equation holds for all $i \ge 0$ (with $p_{-1} = 0$), and at $i = 0$ it reads $\dfrac{\mathrm{d}p_0}{\mathrm{d}t} = \mu p_1$ — internal extinction by death still feeds state 0, while ruptures leave the count altogether.

The derivation proceeds by manipulating the equations to produce terms containing the probability generating function, $G(z,t) = \sum_{i=0}^{\infty} p_i z^i$. It begins by multiplying each side of every ith equality by z^i. In the case of $i = 0$, $z^0 = 1$, so there is no change.

$$\begin{cases} \dfrac{\mathrm{d}p_i z^i}{\mathrm{d}t} = \lambda(i-1)p_{i-1}z^i + \mu(i+1)p_{i+1}z^i - (\lambda+\mu+\delta)ip_i z^i, \quad i > 0 \\[2ex] \dfrac{\mathrm{d}p_0}{\mathrm{d}t} = \mu p_1 + \delta J(t). \end{cases}$$

Then, each equation is combined in a sum over all values of i for which there is a positive term:

$$\sum_{i=0}^{\infty} \frac{\partial p_i z^i}{\partial t} = \delta J(t) + \sum_{i=0}^{\infty} \left[\lambda(i-1)p_{i-1}z^i + \mu(i+1)p_{i+1}z^i - (\lambda+\mu+\delta)ip_i z^i \right]$$

Some index-shifting converts this into

$$\frac{\partial G(z,t)}{\partial t} = \delta J(t) + \lambda z^2 \sum_{i=1}^{\infty} ip_i z^{i-1} + \mu \sum_{i=1}^{\infty} ip_i z^{i-1} - (\lambda+\mu+\delta) z \sum_{i=1}^{\infty} ip_i z^{i-1}$$

and since

$$\sum_{i=1}^{\infty} ip_i z^{i-1} = \frac{\partial}{\partial z} \sum_{i=1}^{\infty} p_i z^i = \frac{\partial G(z,t)}{\partial z},$$

the above becomes

$$\frac{\partial G(z,t)}{\partial t} = \delta J(t) + \lambda z^2 \frac{\partial G}{\partial z} + \mu \frac{\partial G}{\partial z} - (\lambda + \mu + \delta) z \frac{\partial G}{\partial z},$$

equivalent to (15). Dropping the $\delta J(t)$ term — no rupture flux enters any state of the count in the catastrophe convention — leaves (14).

4 Identifying the PGFs

G(z,t), the probability generating function, is a powerful tool for understanding the behaviour of the system and its probable outcomes. Because it can be written as a polynomial sum in which each coefficient is the probability of the process being in a specific state at a given time, valuable information on the process can be extracted.

As,

$$G(z,t) = p_0(t) + p_1(t)z + p_2(t)z^2 + p_3(t)z^3 + \dots,$$

The mean, $J(t)$ can be found with $\partial_z G(z,t)|_{z=1}$:

$$\mathbb{E}(\mathbf{X}_t) = p_1(t) + 2p_2(t) + 3p_3(t) + \dots;$$

Individual state-probabilities can be extracted in the following way. Differentiate with respect to z, n times,

$$\frac{\partial^n G(z,t)}{\partial z^n} = n! \cdot p_n(t) + (n+1)! \cdot p_{n+1}(t)z + \frac{(n+2)!}{2} \cdot p_{n+2}(t)z^2 + \dots,$$

then evaluate at $z = 0$ and divide by $n!$:

$$\frac{1}{n!} \cdot \frac{\partial^n G(z,t)}{\partial z^n}\bigg|_{z=0} = p_n(t). \tag{16}$$

In this way the full dynamic distribution of the system can be elucidated.

Though to do this, we must first obtain the explicit formula $G(z,t)$. There will be some difference between the two cases, but the process is similar — both (14) and (15) can be solved with the method of characteristics.

4.1 Method of characteristics

Generally, first order, linear PDEs of the type we are dealing can be solved as a family of ODEs. The method begins by introducing a change of variables $(z,t) \to (s,\tau)$;

$$G(z,t) \equiv G\left(z(s,\tau), t(s,\tau)\right) = G(s,\tau).$$

It proceeds by seeking solutions on characteristic lines of constant s: $G(\tau)$. An initial curve must be fixed, from which these solutions will progress. The selection is somewhat arbitrary; many options will work, though a good choice can keep the analysis more simple. There are some hard conditions. For example, the initial line must not be parallel in the s, τ space to the progressing solutions. In cases such as ours, it is often appropriate to use

$$z(s,0) = s \qquad\qquad t(s,0) = 0 \tag{17}$$

Catastrophe

Assuming the probability generating function changes with only one variable — $G(\tau(z,t))$ — we can write down the characteristic equations as ODEs. First, consider

$$\frac{\mathrm{d}G}{\mathrm{d}\tau} = \left\{\frac{\mathrm{d}t}{\mathrm{d}\tau}\right\}\frac{\partial G}{\partial t} + \left\{\frac{\mathrm{d}z}{\mathrm{d}\tau}\right\}\frac{\partial G}{\partial z}.$$

Comparing with the catastrophe equation, (14), there is a similarity,

$$0 = \{-1\}\frac{\partial G}{\partial t} + \left\{\lambda z^2 - (\lambda+\mu+\delta)z + \mu\right\}\frac{\partial G}{\partial z},$$

and we can extract the three characteristic ODEs:

$$\frac{\mathrm{d}G}{\mathrm{d}\tau} = 0, \qquad \frac{\mathrm{d}t}{\mathrm{d}\tau} = -1, \qquad \frac{\mathrm{d}z}{\mathrm{d}\tau} = \lambda z^2 - (\lambda+\mu+\delta)z + \mu.$$

The first of these, which will differ in the killing case, tells us that

$$G = \phi(s),$$

ϕ constant on curves of unchanging s.

In both cases, the initial condition on G will be

$$G(z,0) = z$$

because starting with a single individual means $p_1(0) = 1$ and $p_{i\neq 1}(0) = 0$. Therefore, $G = \phi(s)$ in conjunction with our choice of initial curve, $z(s,0) = s$, allows $\phi(s)$ to be fixed as s;

$$G(s,\tau) = s.$$

Moving to the second characteristic ODE:

$$\frac{\mathrm{d}t}{\mathrm{d}\tau} = -1 \quad\Longrightarrow\quad t = -\tau + C_1(s) \text{ (constant for unchanging } s\text{)}.$$

Our initial condition, $t(s,0) = 0$ fixes $C_1(s) = 0$, so $t = -\tau$.

The remaining characteristic equation with z happens to be the same form as an ODE describing the no-rupture probability, $I(t)$.

$$\frac{\mathrm{d}z}{\mathrm{d}\tau} = \lambda z^2 - (\lambda+\mu+\delta)z + \mu \tag{18}$$

What differs is the initial condition. In the case of, $I(t)$, at time 0 it is 1 — rupture having definitely not occurred; with $z(s,t)$, our chosen initial condition is $z(s,0) = s$. Hence, the solution to (18) is very similar to the formula for $I(t)$, though in places where there is seen a one, here an s is found:

$$z = \frac{a(s-b) - b(s-a)\mathrm{e}^{\lambda(a-b)\tau}}{(s-b) - (s-a)\mathrm{e}^{\lambda(a-b)\tau}}$$

Also, in this it is τ rather than t.

Since $G(z,t) = s(z,t)$, we can obtain the final solution in one step: Rearrange the above for s, and place $-t$ at each τ. By a happenstance of symmetry, this turns out looking quite the same:

$$s = \frac{a(z-b) - b(z-a)\mathrm{e}^{\lambda(a-b)t}}{(z-b) - (z-a)\mathrm{e}^{\lambda(a-b)t}}. \tag{19}$$

And so we have it, by the somewhat dark art of characteristics, the probability generating function emerges as

$$G(z,t) = \frac{a(z-b) - b(z-a)\mathrm{e}^{\lambda(a-b)t}}{(z-b) - (z-a)\mathrm{e}^{\lambda(a-b)t}}.$$

As will be seen, in obtaining the state probabilities via repeated partial differentiation in z, it is advantageous to write the formula as

$$G(z,t) = \frac{ab(1-\mathrm{e}^{\lambda(a-b)t}) - z(a - b\mathrm{e}^{\lambda(a-b)t})}{(b - a\mathrm{e}^{\lambda(a-b)t}) - z(1-\mathrm{e}^{\lambda(a-b)t})}$$

For now, let us move on to the modification that is introduced in the killing case.

Killing

When rupture collapses the process to 0, it may have happened at any positive state. As we've seen, because of this, the forward Kolmogorov equation describing $p_0(t)$ includes a separate term related to every natural number. Thankfully, these do compress neatly into an expression of the mean, $\mathbb{E}(\mathbf{X}_t) = J(t)$. Hence, in the PDE, (15), there is an inhomogeneous term: $\delta J(t)$.

Making the same comparison as in the catastrophe case,

$$\begin{aligned} \frac{\mathrm{d}G}{\mathrm{d}\tau} &= \left\{\frac{\mathrm{d}t}{\mathrm{d}\tau}\right\}\frac{\partial G}{\partial t} + \left\{\frac{\mathrm{d}z}{\mathrm{d}\tau}\right\}\frac{\partial G}{\partial z} \\ -\delta J(t) &= \{-1\}\frac{\partial G}{\partial t} + \left\{\lambda z^2 - (\lambda+\mu+\delta)z + \mu\right\}\frac{\partial G}{\partial z} \end{aligned}$$

we see the characteristic equations are largely the same, just that the first picks up this extra component:

$$\frac{\mathrm{d}G}{\mathrm{d}\tau} = -\delta J(-\tau), \qquad \frac{\mathrm{d}t}{\mathrm{d}\tau} = -1, \qquad \frac{\mathrm{d}z}{\mathrm{d}\tau} = \lambda z^2 - (\lambda+\mu+\delta)z + \mu.$$

From previous work, we know

$$-\delta J(t) = \frac{\mathrm{d}I(t)}{\mathrm{d}t}.$$

Simply because $\mathrm{d}t = -\mathrm{d}\tau$, it follows that

$$\delta J(-\tau) = \frac{\mathrm{d}I(-\tau)}{\mathrm{d}\tau}.$$

And so, looking to the first characteristic equation,

$$\frac{\mathrm{d}G}{\mathrm{d}\tau} = \frac{\mathrm{d}I(-\tau)}{\mathrm{d}\tau},$$

hence,

$$G(s,\tau) = -I(-\tau) + \phi(s).$$

Again, with some ϕ stationary for over constant s.

With the initial condition, $G(z, t=0) = z(s, \tau=0) = s$,

$$G(s,0) = s = -I(0) + \phi(s),$$

ϕ can be fixed: $\phi(s) = 1 + s$;

$$G(s, \tau) = 1 - I(-\tau) + s.$$

Identical to the catastrophe case, the final characteristic equation is solved in the same way for the same result, (19). So finally,

$$G(z,t) = 1 - I(t) + \frac{ab(1 - \mathrm{e}^{\lambda(a-b)t}) - z(a - b\mathrm{e}^{\lambda(a-b)t})}{(b - a\mathrm{e}^{\lambda(a-b)t}) - z(1 - \mathrm{e}^{\lambda(a-b)t})}.$$

And this result makes good sense. With $1 - I(t)$ being the likelihood that the process has made the "killing" transition to 0, these two terms stand to complement the probability that state 0 is reached from individual death (this component arises from the final term).

4.2 Several founders

Both solutions extend to an arbitrary initial condition by the branching property. If the cell begins with k independent founders, $X_0 = k$, then the k lineages evolve independently and the total count is their sum, so the probability generating function is the kth power of the single-founder probability generating function:

$$G_k(z,t) = \left(G(z,t)\right)^k, \tag{20}$$

in either rupture convention. In the catastrophe convention this is a defective probability generating function with $G_k(1,t) = I(t)^k$; in the killing convention it remains proper. Formula (20) will be used in §9 (multiplicity of infection) and §10 (chained immediate transfer).

5 State probabilities

Using formula (16), we can obtain the individual state probabilities. The first, $p_0(t)$, differs between the catastrophe and killing cases:

$$G(0,t) = \begin{cases} \frac{ab(1-\mathrm{e}^{\lambda(a-b)t})}{(b-a\mathrm{e}^{\lambda(a-b)t})}, & \text{(catastrophe)} \\ 1 - I(t) + \frac{ab(1-\mathrm{e}^{\lambda(a-b)t})}{(b-a\mathrm{e}^{\lambda(a-b)t})}, & \text{(killing).} \end{cases}$$

As stated above, the fraction term in each case represents the probability of reaching 0 by the death of every individual, and the $1 - I(t)$ pertains to reaching 0 by collapse in the killing convention.

From here, the state probabilities $p_1(t)$, $p_2(t)$, $p_3(t)\dots$ do not differ between the cases. Repeated differentiation of $G(z,t)$ with respect to z reveals the formula,

$$\frac{\partial^n G}{\partial z^n} = n!\mathrm{e}^{\lambda(a-b)t}(a-b)^2 \frac{\left(1 - \mathrm{e}^{\lambda(a-b)t}\right)^{n-1}}{\left(b - a\mathrm{e}^{\lambda(a-b)t} - \left(1 - \mathrm{e}^{\lambda(a-b)t}\right) z\right)^{n+1}}$$

Hence, setting $z = 0$, and dividing by $n!$, the state probabilities for $n \geq 1$ are described by,

$$p_n(t) = \left(\frac{a-b}{b - a\mathrm{e}^{\lambda(a-b)t}}\right)^2 \mathrm{e}^{\lambda(a-b)t} \left(\frac{1 - \mathrm{e}^{\lambda(a-b)t}}{b - a\mathrm{e}^{\lambda(a-b)t}}\right)^{n-1}. \tag{21}$$

5.1 Geometric at every time

There is more structure here than (21) first suggests. Writing

$$P(t) := \frac{1 - \mathrm{e}^{\lambda(a-b)t}}{b - a\mathrm{e}^{\lambda(a-b)t}}, \tag{22}$$

the entire n-dependence of $p_n(t)$ lives in the single factor $P(t)^{n-1}$:

$$p_n(t) = p_1(t)\, P(t)^{n-1}, \qquad n \geq 1. \tag{23}$$

That is, *at every time* t, conditional on the population being non-zero, the load is geometrically distributed with ratio $P(t)$. The ratio itself evolves monotonically, with

$$P(0) = 0, \qquad P(t) \xrightarrow[t\to\infty]{} \frac{1}{a}, \tag{24}$$

so the transient geometric laws slide towards a fixed limit. This limit is the heart of §6–§7.

5.2 Normalisation: the sum is I_{fix}

A useful consistency check is to sum (21) over the positive states. As a geometric series,

$$\sum_{n=1}^{\infty} p_n(t) = \frac{p_1(t)}{1 - P(t)}.$$

Now

$$1 - P(t) = \frac{b - a\mathrm{e}^{\lambda(a-b)t} - 1 + \mathrm{e}^{\lambda(a-b)t}}{b - a\mathrm{e}^{\lambda(a-b)t}} = \frac{-(1-b) - \mathrm{e}^{\lambda(a-b)t}(a-1)}{b - a\mathrm{e}^{\lambda(a-b)t}} = \frac{B + A\mathrm{e}^{\lambda(a-b)t}}{a\mathrm{e}^{\lambda(a-b)t} - b},$$

and therefore

$$\sum_{n=1}^{\infty} p_n(t) = \frac{(a-b)^2 \mathrm{e}^{\lambda(a-b)t}}{\left(B + A\mathrm{e}^{\lambda(a-b)t}\right)\left(a\mathrm{e}^{\lambda(a-b)t} - b\right)} = I_{\text{fix}}(t). \tag{25}$$

The sum of the positive state probabilities is exactly the non-fixation probability of Section 2.4 — the probability of being neither ruptured nor extinct. Equation (25) thereby confirms, by a completely independent route, the closed form (7): the state probabilities of this section imply the same $I_{\text{fix}}(t)$ that Chapter 3's decomposition produced.

FIGURE FLAG F4a.2 — Geometric slide of the state probabilities
Purpose: make the section's central fact visible — that the load is geometrically distributed at *every* time, with a ratio $P(t)$ that slides monotonically to $1/a$. **Type:** Python analytic plot. **Content:** $p_n(t)$ against $n = 1, \ldots, 60$ for $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$ at the five times $t \in \{0.5, 1, 2, 5, 15\}$, evaluated from the closed form (21) with $w = e^{\theta t}$, $\theta = \lambda(a-b)$; annotate each curve with its value of $P(t) = (1-w)/(b-aw)$, and mark the limit $1/a$ (e.g. via an inset plotting $P(t)$ against t from (22)). **Axes:** n (horizontal), $p_n(t)$ (vertical, log scale optional); legend entries $t = 0.5, 1, 2, 5, 15$. **Style:** unified Python style — default matplotlib with `tab:` colours, grid alpha 0.25, `tight_layout`, PDF output, fonts matching the chapter. **Data source:** analytic formula (21); no simulation. **Placement:** Section 5, after the "Geometric at every time" subsection.

6 Quasi-stationary distribution

[4]. Both the BDC and BDK processes are subject to eventual absorption — either at the designated rupture state, or by death, at 0. But suppose there is an example of a particular realisation which has, by chance, not yet reached fixation. It continues, increasingly against the odds, as time marches on. How has it managed this? If it flies too low, depletion to zero becomes more and more probable; but, almost as bad, if it flies high, a population booming in number, there are all the more opportunities for its downfall, a crashing and ceasing to be. Like Icarus, the process which does not fall into the sea does so by hovering at some appropriate height; its wings neither undermined by the spray of a hungry ocean, nor by the beating rays of the sun at a rarified altitude.

In such terms, we can examine this rogue process as it goes on to defy all odds. Its fate will tend to be predicted by the quasi-stationary distribution: a set of limiting conditional probabilities, each qualified by non-fixation.

6.1 Conditioning on survival

Define the probability of fixation by time t: $F(t)$, whether by death or collapse. This is settled immediately by the decomposition of Section 2.4: a realisation is fixed by t precisely when it is not productively infected, so

$$F(t) = 1 - I_{\text{fix}}(t) = 1 - I(t) + D(t), \tag{26}$$

which reduces to $F(t) = 1 - I(t)$ when $\mu = 0$. The conditional state probabilities decompose as

$$\Pr\{\mathbf{X}_t = i\} = \Pr\{\mathbf{X}_t = i \mid \text{fixation}\} F(t) + \Pr\{\mathbf{X}_t = i \mid \text{non-fixation}\}\,(1 - F(t))\,.$$

For $i \notin \{0, H\}$ the process has not been absorbed, so $\Pr\{\mathbf{X}_t = i \mid \text{fixation}\} = 0$, and it follows that

$$\Pr\{\mathbf{X}_t = i \mid \text{non-fixation}\} = \frac{p_i(t)}{1 - F(t)} = \frac{p_i(t)}{I_{\text{fix}}(t)}. \tag{27}$$

The *quasi-stationary distribution* (QSD) is the limiting form of these conditional probabilities:

$$q_i := \lim_{t\to\infty} \Pr\{\mathbf{X}_t = i \mid \mathbf{X}_t \notin \{0, H\}\} = \lim_{t\to\infty} \frac{p_i(t)}{I_{\text{fix}}(t)}, \qquad i \geq 1. \tag{28}$$

6.2 The quasi-stationary distribution is geometric

With the state probabilities (21) in hand, the limit can be taken explicitly. By (27) and (25),

$$\Pr\{\mathbf{X}_t = n \mid \mathbf{X}_t \notin \{0, H\}\} = \frac{p_n(t)}{\sum_{m\geq 1} p_m(t)} = (1 - P(t))\, P(t)^{n-1},$$

for every t. Since $P(t) \to 1/a$ by §5.1, the limit exists and is explicit.

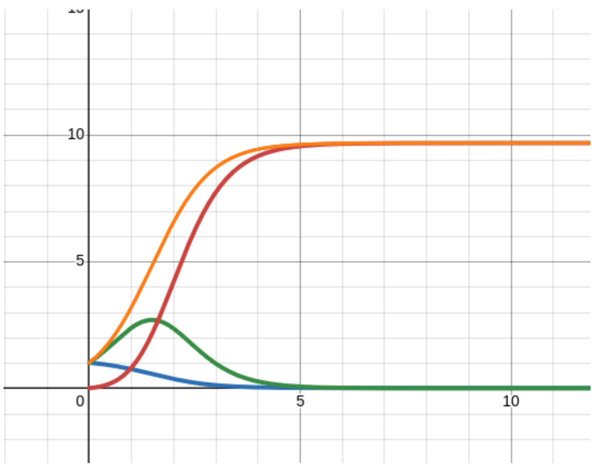
Theorem 6.1 (Quasi-stationary distribution of the BDC)**.** *For all $\mu \geq 0$, the quasi-stationary distribution of the BDC process is geometric on $\{1, 2, \ldots\}$ with ratio $1/a$:*

$$q_n = \left(1 - \frac{1}{a}\right)\left(\frac{1}{a}\right)^{n-1} = (a-1)\, a^{-n}, \qquad n \geq 1. \tag{29}$$

Its moments follow at once:

$$\langle X\rangle_{\mathrm{QS}} = \frac{a}{a-1} = \frac{a}{A}, \qquad \langle X^2\rangle_{\mathrm{QS}} = \frac{a(a+1)}{(a-1)^2}, \qquad \mathrm{Var}_{\mathrm{QS}}(X) = \frac{a}{(a-1)^2} = \frac{a}{A^2}. \tag{30}$$

Observe that $\langle X\rangle_{\mathrm{QS}} = \mathbb{E}(\mathcal{K} \mid \text{burst})$ from (12), and that when $\mu = 0$ both coincide with V_∞ — the coincidence noted in §2.5. For $\mu > 0$ they differ: at $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$, for example, $\langle X\rangle_{\mathrm{QS}} \approx 17.23$ while $V_\infty \approx 13.99$. Section 7 will give the reason: the burst size is not merely equal in mean to the QS level — the entire burst-size distribution *is* the quasi-stationary distribution.



(a) When $\mu = 0$.

(b) When $\mu > 0$.

Figure 1: Conditional means from simulation. (a) When $\mu = 0$, the conditional mean $J(t)/I(t)$ tends to the same limit as $V(t)$, namely $1 + \lambda/\delta$. (b) When $\mu > 0$, the curve rising at late time is the mean conditioned on non-fixation, $J(t)/I_{\mathrm{fix}}(t)$, tending to $\langle X\rangle_{\mathrm{QS}} = a/(a-1)$; the curve tending to 0 is the mean conditioned only on non-catastrophe, $\mathbb{E}(\mathbf{X}_t \mid \mathbf{X}_t \neq H) = J(t)/I(t)$, which vanishes because a fraction b of realisations die out internally.

FIGURE FLAG F4a.3 — Convergence to the QSD

Purpose: show the quasi-stationary theorem numerically — the conditional pmf $p_n(t)/I_{\mathrm{fix}}(t)$ converging, as t grows, to the limiting geometric law $(a-1)a^{-n}$ of Theorem 6.1. **Type:** Python analytic plot (optionally with a simulated conditional histogram overlay). **Content:** for $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$: curves of the conditional pmf $p_n(t)/I_{\mathrm{fix}}(t)$ against $n = 1, \ldots, 60$ at $t \in \{1, 5, 20\}$, evaluated from (21) and (7); overlay the limiting geometric pmf $(a-1)a^{-n}$ as a dashed curve; optionally add a Gillespie-simulated conditional histogram at $t = 20$ (ensemble $\geq 10^5$ realisations, conditioning on $X_t \notin \{0, H\}$) to demonstrate agreement. **Axes:** n (horizontal), conditional probability (vertical); legend entries $t = 1, 5, 20$ and "limit $(a-1)a^{-n}$". **Style:** unified Python style — default matplotlib with `tab:` colours, grid alpha 0.25, `tight_layout`, PDF output, fonts matching the chapter. **Data source:** analytic formulas (21), (7); simulation recipe (optional): exact Gillespie with rates λn, μn, δn. **Placement:** Section 6, directly after Figure 1.

Considering the no-death $\mu = 0$ case on the left above, it can be easily seen that the conditional mean tends to the same limit as $V(t)$, which is shown in red. That is

$$\lim_{t\to\infty} \frac{J(t)}{I(t)} = \lim_{t\to\infty} V(t) = 1 + \frac{\lambda}{\delta}.$$

Here, when δ shrinks to 0, the QS mean and expected release count tend to infinity.

6.3 Mean productive lifetime

Before leaving the single-cell description, one more integral will be needed when, in Chapter 4b, the single-cell theory is embedded in a population model. The expected duration of productive infection is

$$\mathbb{E}(T_{\mathrm{prod}}) = \int_0^\infty I_{\mathrm{fix}}(t)\,\mathrm{d}t, \tag{31}$$

since $I_{\mathrm{fix}}(t) = \mathbb{P}\{T_{\mathrm{prod}} > t\}$.

Proposition 6.2.

$$\mathbb{E}(T_{\mathrm{prod}}) = \frac{1}{\lambda}\log\Big(\frac{a}{a-1}\Big) = \frac{1}{\lambda}\log\langle X\rangle_{\mathrm{QS}}. \tag{32}$$

Proof. Write $I_{\mathrm{fix}} = (I-b) - (D-b)$. From (7),

$$I(t) - b = \frac{B(a-b)}{B+Aw}, \qquad D(t) - b = -\frac{b(a-b)}{aw-b},$$

so that, with $w = \mathrm{e}^{\theta t}$ and $\mathrm{d}t = \mathrm{d}w/(\theta w)$,

$$\int_0^\infty I_{\mathrm{fix}}\,\mathrm{d}t = \frac{a-b}{\theta}\int_1^\infty \left(\frac{B}{w(B+Aw)} + \frac{b}{w(aw-b)}\right)\mathrm{d}w.$$

By partial fractions, $\frac{B}{w(B+Aw)} = \frac{1}{w} - \frac{A}{B+Aw}$ and $\frac{b}{w(aw-b)} = -\frac{1}{w} + \frac{a}{aw-b}$, hence

$$\int_1^\infty \frac{B\,\mathrm{d}w}{w(B+Aw)} = \Big[\log\frac{w}{B+Aw}\Big]_1^\infty = \log\frac{a-b}{a-1},$$

$$\int_1^\infty \frac{b\,\mathrm{d}w}{w(aw-b)} = \Big[\log\frac{aw-b}{w}\Big]_1^\infty = \log\frac{a}{a-b}.$$

Summing, and using $\theta = \lambda(a-b)$,

$$\int_0^\infty I_{\mathrm{fix}}\,\mathrm{d}t = \frac{a-b}{\lambda(a-b)}\left(\log\frac{a-b}{a-1} + \log\frac{a}{a-b}\right) = \frac{1}{\lambda}\log\frac{a}{a-1}. \qquad \square$$

$\square$

When $\mu = 0$ this reduces to $\lambda^{-1}\log(1+\lambda/\delta)$. The form of the answer has an interpretation worth recording: $\log\langle X\rangle_{\mathrm{QS}}/\lambda$ is the time it would take exponential growth at rate λ to reach the quasi-stationary level — the productive lifetime is precisely the time needed to climb to the height at which the process can hover.

7 Burst time and burst size

The two random quantities that a bursting cell presents to the outside world are the time of its rupture and the size of the rupture. Both are now within reach of closed forms.

7.1 Burst time

The no-burst probability is $I(t) = \mathbb{P}\{\tau > t\}$, so the density of the burst time is

$$\varphi(t) := -I'(t) = \delta J(t), \tag{33}$$

using $I' = -\delta J$ from §2.5. This density is *defective*: since $I(t) \to b$,

$$\int_0^\infty \varphi(t)\,\mathrm{d}t = 1 - b, \tag{34}$$

with the remaining mass b sitting at $\tau = \infty$ — the realisations that clear internally without ever bursting. Conditioned on eventual burst, the density is $\varphi(t)/(1-b) = \delta J(t)/(1-b)$. Where earlier work referred to "the known pdf of rupture times $\phi(t)$", it is (33) that was meant, available all along from $I' = -\delta J$.

FIGURE FLAG F4a.4 — Burst-time density

Purpose: display the defective burst-time density, making the mass b at $\tau = \infty$ and the conditional density explicit in one picture. **Type:** Python analytic plot. **Content:** two parameter sets, $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$ and $(1, 0, 0.1)$, on $t \in [0, 15]$: the defective density $\varphi(t) = \delta J(t)$ from (33) and (8) (solid curves); for the $\mu > 0$ set, shade and annotate the missing mass b at ∞ (e.g. an annotated bar or hatched region labelled "mass b at $\tau = \infty$"); overlay the conditional density $\delta J(t)/(1-b)$ as a dashed curve for the $\mu > 0$ set. **Axes:** t (horizontal), density (vertical); legend entries $\varphi(t)$ and $\varphi(t)/(1-b)$ per parameter set. **Style:** unified Python style — default matplotlib with `tab:` colours, grid alpha 0.25, `tight_layout`, PDF output, fonts matching the chapter. **Data source:** analytic formulas (33), (8); no simulation. **Placement:** Section 7.1, directly after the defect discussion.

7.2 Burst size

The burst size $\mathcal{K}$ is the value of $\mathbf{X}_t$ immediately before rupture. For the burst to occur in state k, the process must be in state k and then rupture from there; rupturing from state k happens at rate δk. Hence

$$\Pr\{\mathcal{K} = k\} = \delta k \int_0^\infty p_k(t)\,\mathrm{d}t. \tag{35}$$

A direct discretisation makes this plain: partitioning time into intervals of width Δt, the probability that the burst occurs during an interval in which the process is at k is

$$\Pr\{\mathbf{X}_t = k\}\Pr\{\text{rupture in } \Delta t \mid \mathbf{X}_t = k\} = \Pr\{\mathbf{X}_t = k\}\,\delta k\,\Delta t,$$

and summing over intervals and passing to the limit $\Delta t \to 0$ yields (35).

The integral is evaluable in closed form. With $P(t)$ as in (22),

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big[\frac{P(t)^k}{k\lambda}\Big] = \frac{1}{\lambda}P(t)^{k-1}P'(t) = \Big(\frac{a-b}{b - a\mathrm{e}^{\lambda(a-b)t}}\Big)^2 \mathrm{e}^{\lambda(a-b)t}P(t)^{k-1} = p_k(t), \tag{36}$$

as one verifies from $P'(t) = \lambda(a-b)^2\mathrm{e}^{\lambda(a-b)t}/(b - a\mathrm{e}^{\lambda(a-b)t})^2$. Therefore, since $P(0) = 0$ and $P(\infty) = 1/a$,

$$\int_0^\infty p_k(t)\,\mathrm{d}t = \frac{1}{k\lambda}\Big(\frac{1}{a}\Big)^k, \tag{37}$$

and (35) gives the burst-size law.

Theorem 7.1 (Burst-size distribution). *Unconditionally (including the mass b of realisations that never burst),*

$$Pr\{\mathcal{K} = k\} = \frac{\delta}{\lambda}\, a^{-k}, \qquad k \geq 1; \tag{38}$$

conditioned on the burst occurring,

$$Pr\{\mathcal{K} = k \mid burst\} = (a-1)\, a^{-k}, \qquad k \geq 1. \tag{39}$$

Proof. The unconditional form is the computation above. For the conditional form, divide (38) by the eventual-burst probability $1 - b$:

$$\Pr\{\mathcal{K} = k \mid \text{burst}\} = \frac{(\delta/\lambda)\, a^{-k}}{1-b} = \frac{AB}{B}\, a^{-k} = (a-1)\, a^{-k},$$

using $\delta/\lambda = AB$ from (3). □

No simulation is required; nevertheless two consistency checks close the loop pleasantly. First, normalisation:

$$\sum_{k=1}^{\infty} \frac{\delta}{\lambda} a^{-k} = \frac{\delta}{\lambda}\, \frac{1}{a-1} = \frac{AB}{A} = B = 1 - b,$$

matching the eventual-burst probability. Second, the first moment:

$$\sum_{k=1}^{\infty} k\, \frac{\delta}{\lambda} a^{-k} = \frac{\delta}{\lambda}\, \frac{a}{(a-1)^2} = \frac{aB}{A} = V_\infty,$$

recovering (12). This second check independently validates the burst-size law, the formula for V_∞, and the identity (3) simultaneously.

7.3 Burst size equals the quasi-stationary distribution

Compare (39) with Theorem 6.1: they are the same law.

Corollary 7.2 (Burst size = QSD). *The burst-size distribution, conditioned on bursting, is exactly the quasi-stationary distribution of the load: both are geometric on $\{1, 2, \ldots\}$ with ratio $1/a$.*

This is not obvious, and it deserves a moment's pause. A burst is a *size-biased* draw from the transient load distribution, taken against the burst intensity, and then integrated over all burst times; that this should land exactly on the conditional limiting distribution is a genuine identity, not a definition. The algebra above proves it; a structural argument — presumably relating size-biasing composed with the burst intensity to the Yaglom limit [5] — would make it quotable rather than coincidental, and remains open (see §11.3).

7.4 Size-biasing and the size of a late burst

The size-biasing mechanism is worth stating on its own terms. A cell bursts at rate $\delta \mathbf{X}_t$, so given that the burst occurs at time t, its size is a size-biased draw from $p.(t)$:

$$\Pr\{\mathcal{K} = k \mid \tau = t\} = \frac{k\, p_k(t)}{J(t)}, \qquad \mathbb{E}(\mathcal{K} \mid \tau = t) = \frac{K(t)}{J(t)} = 1 + \frac{2\lambda}{\delta}\,(1 - I(t))\,. \tag{40}$$

Taking $t \to \infty$,

$$\mathbb{E}(\mathcal{K} \mid \tau = t) \xrightarrow[t\to\infty]{} 1 + \frac{2\lambda}{\delta}(1-b) = \frac{\langle X^2\rangle_{\mathrm{QS}}}{\langle X\rangle_{\mathrm{QS}}}, \tag{41}$$

where the second equality follows from (30). A very late burst is roughly twice the size of an average one — the signature of size-biasing a geometric law: the cells that survive long are precisely the ones that grew large, and large cells burst quickly.

FIGURE FLAG F4a.5 — Burst size and late bursts

Purpose: show the two identities of this section together — the burst-size law, and the size-biasing effect by which late bursts exceed average ones. **Type:** Python analytic plot, two panels. **Content:** panel (a): bars of the unconditional burst-size law $\Pr\{\mathcal{K} = k\} = (\delta/\lambda)a^{-k}$, $k = 1, \ldots, 40$, for $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$, with the conditional law $(a-1)a^{-k}$ overlaid as a connected curve, and the mass b at $k = 0$ annotated (bar or text). Panel (b): the size-biased conditional mean $K(t)/J(t) = 1 + \frac{2\lambda}{\delta}(1 - I(t))$ against $t \in [0, 25]$ for the same parameters, with the horizontal asymptote $(a+1)/(a-1) \approx 33.5$ dashed and annotated; optionally overlay the QS mean $a/(a-1)$ for contrast. **Axes:** panel (a): burst size k (horizontal), probability (vertical); panel (b): t (horizontal), $\mathbb{E}(\mathcal{K} \mid \tau = t)$ (vertical). **Style:** unified Python style — default matplotlib with `tab:` colours, grid alpha 0.25, `tight_layout`, PDF output, fonts matching the chapter. **Data source:** analytic formulas (38), (39), (40); no simulation. **Placement:** Section 7.4, after the size-biasing discussion.

7.5 Burst time given burst size, without death

Size-biasing conditions the burst size on the burst time; the converse conditioning — the burst time given the burst size — has an equally explicit form when $\mu = 0$, and it is the quantity an experiment accesses when it measures both members of each (burst time, burst size) pair. Write $\theta = \lambda + \delta$ for the rate combination that governs the $\mu = 0$ case (it is $\lambda(a - b)$ evaluated at $b = 0$). Fixing n and regarding $p_n(t)$ as a function of t, the state probability rises from zero, peaks, and decays; differentiating $\log p_n(t)$ shows the maximum is attained at

$$t_n = \frac{\log n}{\theta} : \tag{42}$$

a burst of size n is most likely to have occurred once the process has had of order $(\log n)/\theta$ of growth time. The exact conditional mean burst time is the following.

Proposition 7.3 (Conditional rupture time, $\mu = 0$)**.** *With $\mu = 0$ and $\theta = \lambda + \delta$,*

$$\mathbb{E}(\tau \mid \mathcal{K} = n) = \frac{1}{\theta}\sum_{k=1}^{n}\frac{1}{k}, \qquad n \geq 1. \tag{43}$$

Proof. When $\mu = 0$ the state probabilities (21) reduce, with $r = \lambda/\delta$ and $s = \theta t$, to

$$p_n(t) = \mathrm{e}^{-\theta t}\left(\frac{1 - \mathrm{e}^{-\theta t}}{1 + 1/r}\right)^{n-1},$$

and the burst-size law (38) to $\Pr\{\mathcal{K} = n\} = (1/r)(r/(r+1))^n$. Writing

$$L_1(n) = \int_0^\infty \mathrm{e}^{-s}(1 - \mathrm{e}^{-s})^{n-1}\,\mathrm{d}s, \qquad L_2(n) = \int_0^\infty s\,\mathrm{e}^{-s}(1 - \mathrm{e}^{-s})^{n-1}\,\mathrm{d}s,$$

the substitution $s = \theta t$ gives $\mathbb{E}(\tau \mid \mathcal{K} = n) = L_2(n)/(\theta L_1(n))$. The first integral is $L_1(n) = 1/n$, by the substitution $u = 1 - \mathrm{e}^{-s}$. For the second, split off one factor of $(1 - \mathrm{e}^{-s})$ to write

$L_2(n+1) = L_2(n) - \int_0^\infty s\mathrm{e}^{-2s}(1-\mathrm{e}^{-s})^{n-1}\mathrm{d}s$, and integrate by parts in the remaining integral; the result is the recurrence

$$(n+1)L_2(n+1) = nL_2(n) + \frac{1}{n+1}, \qquad L_2(1) = 1,$$

whose solution is $L_2(n) = \frac{1}{n}(1 + \frac{1}{2} + \cdots + \frac{1}{n})$; the quotient is (43). $\square$

For large n the harmonic sum gives the approximation

$$\mathbb{E}(\tau \mid \mathcal{K} = n) \simeq \frac{\log n + \gamma}{\theta} \qquad (n \gg 1), \tag{44}$$

where $\gamma = 0.5772\ldots$ is Euler's constant: conditional rupture time grows logarithmically with burst size. The full conditional distribution has a clean form in the same limit. Since the density of τ at t, given $\mathcal{K} = n$, is proportional to $p_n(t)$,

$$\Pr\{\tau < t \mid \mathcal{K} = n\} = \frac{\delta n \int_0^t p_n(s)\,\mathrm{d}s}{\Pr\{\mathcal{K} = n\}} = (1 - \mathrm{e}^{-\theta t})^n, \tag{45}$$

and for $n \gg 1$, with $T = \theta(t - t_n)$ and t_n as in (42),

$$\Pr\{\tau < t \mid \mathcal{K} = n\} \simeq \exp\left(-n\mathrm{e}^{-\theta t}\right) = \exp\left(-\mathrm{e}^{-T}\right): \tag{46}$$

the conditioned rupture time, centred at t_n and scaled by $1/\theta$, converges to the standard Gumbel law. Burst sizes thus select burst times from a travelling window of width of order $1/\theta$ around $(\log n)/\theta$.

7.6 The mean burst time when death is possible

When $\mu > 0$ the burst time τ has no finite mean, because the mass b at $\tau = \infty$ (the internally cleared cells) makes the integral diverge. Conditioned on bursting, however, the mean is finite and explicit.

Proposition 7.4 (Mean burst time conditioned on bursting)**.** *For $\mu > 0$,*

$$\mathbb{E}(\tau \mid \textit{burst}) = \frac{1}{\lambda(1-b)} \log \frac{a-b}{a-1}. \tag{47}$$

Proof. The density of τ on the bursting realisations is $\varphi(t) = \delta J(t)$, with total mass $1 - b$, so $\mathbb{E}(\tau \mid \text{burst}) = (1-b)^{-1}\int_0^\infty t\,\varphi(t)\,\mathrm{d}t$. Since $\varphi = -(I - b)'$, integrating by parts on $[0, T]$ gives

$$\int_0^T t\,\varphi(t)\,\mathrm{d}t = -T\big(I(T) - b\big) + \int_0^T \big(I(t) - b\big)\,\mathrm{d}t,$$

and the boundary term vanishes as $T \to \infty$ because $I(T) - b = B(a-b)/(B + Aw)$, read off from (7), decays like $\mathrm{e}^{-\theta T}$. It remains to evaluate the integral: with $\mathrm{d}t = \mathrm{d}w/(\theta w)$ and the partial-fraction decomposition $1/[w(B + Aw)] = B^{-1}(1/w - A/(B + Aw))$,

$$\int_0^\infty \big(I(t) - b\big)\,\mathrm{d}t = \frac{B(a-b)}{\theta}\int_1^\infty \frac{\mathrm{d}w}{w(B + Aw)} = \frac{1}{\lambda}\log\frac{a-b}{a-1},$$

and dividing by $1 - b$ gives (47). $\square$

This quantity is distinct from the mean productive lifetime $\mathbb{E}(T_{\text{prod}}) = \lambda^{-1}\log\frac{a}{a-1}$ of §6.3, and the distinction should be kept visible: $\mathbb{E}(\tau \mid \text{burst})$ averages the rupture time over cells that burst, while $\mathbb{E}(T_{\text{prod}})$ averages the time spent productive over *all* cells, the internally cleared ones included with their shorter productive spans. When $\mu = 0$ the two coincide, both reducing to $\lambda^{-1}\log(1 + \lambda/\delta)$, and it is tempting to read more into the coincidence than it supports.

7.7 Comparison with budding release

The bursting single-cell picture just completed has a natural comparator: the budding cell of the constant-rate models. In its simplest stochastic form, a budding infected cell experiences two *independent* event types — release of one infectious particle at rate p, and death of the cell at rate $d_{\mathcal{I}}$ — and the two continue until the cell dies. Its survival and mean release are then

$$I_{\mathrm{bud}}(t) = \mathrm{e}^{-d_{\mathcal{I}}t}, \qquad V_{\mathrm{bud}}(t) = \frac{p}{d_{\mathcal{I}}}\left(1 - \mathrm{e}^{-d_{\mathcal{I}}t}\right),$$

the same one-cell limit that the Basic Model of Viral Replication reaches at the population level. The release count of a budding cell is geometric on $\{0, 1, 2, \ldots\}$:

$$\Pr\{\mathcal{K}_{\mathrm{bud}} = k\} = \frac{d_{\mathcal{I}}}{d_{\mathcal{I}} + p}\left(\frac{p}{d_{\mathcal{I}} + p}\right)^k, \qquad k = 0, 1, 2, \ldots, \tag{48}$$

since each of the successive events is a release with probability $p/(d_{\mathcal{I}} + p)$ and the terminating death with probability $d_{\mathcal{I}}/(d_{\mathcal{I}} + p)$. The structural contrast with bursting is exact: in budding, release and death are independent events, and a cell may release many particles and then die with nothing left to release; in bursting, release and death are the *same* event, and whatever has accumulated is released at once. Table 2 collects the two single-cell descriptions side by side for the $\mu = 0$ bursting case.

	budding	bursting ($\mu = 0$)
$\dfrac{\mathrm{d}I}{\mathrm{d}t}$	$-d_{\mathcal{I}}I$	$-(\lambda + \delta)I + \lambda I^2$
$I(t)$	$\mathrm{e}^{-d_{\mathcal{I}}t}$	$\dfrac{\lambda + \delta}{\lambda + \delta\,\mathrm{e}^{(\lambda+\delta)t}}$
mean cell lifetime	$1/d_{\mathcal{I}}$	$\dfrac{1}{\lambda}\log\left(1 + \dfrac{\lambda}{\delta}\right)$
$V(t)$	$\dfrac{p}{d_{\mathcal{I}}}(1 - I)$	$\dfrac{\lambda}{\delta} + 1 - \dfrac{2\lambda + \delta}{\delta}I + \dfrac{\lambda}{\delta}I^2$
release law	geometric on $\{0, 1, \ldots\}$, (48)	geometric on $\{1, 2, \ldots\}$, $\Pr\{\mathcal{K} = k\} = \frac{\delta}{\lambda}a^{-k}$

Table 2: Single-cell budding and bursting compared, $\mu = 0$ for bursting. Budding parameters: release rate p and infected-cell death rate $d_{\mathcal{I}}$. Bursting parameters: division rate λ and rupture rate δ per intracellular particle; $a = (\lambda + \delta)/\lambda$ here.

Two rows of the table carry the comparison. The mean cell lifetime of a bursting cell grows only logarithmically with λ/δ, while its mean release grows linearly; budding offers no such coupling, its lifetime and its release controlled by separate parameters. And the release laws, both geometric, differ in their support: budding can release nothing at all from a dead cell, while bursting never does. These two differences are what the population-level comparison of Chapter 4b exploits, where bursting's simultaneous release-and-death translates into a measurable advantage at small population sizes.

FIGURE FLAG F4a.8 — Conditional rupture time versus burst size

Purpose: show the harmonic-number law (43) against burst size, with its large-n approximation and the linear budding comparator, in one figure. **Type:** Python analytic plot. **Content:** main panel: $\mu = 0$ with $\theta = \lambda + \delta = 1.1$ (e.g. $\lambda = 1$, $\delta = 0.1$): red dots at the exact means $\mathbb{E}(\tau \mid \mathcal{K} = n) = \theta^{-1} \sum_{k=1}^{n} 1/k$ for $n = 1, \ldots, 40$; dotted curve $\theta^{-1}(\log n + \gamma)$ for $n \geq 2$; dashed straight line $(n+1)/(d_{\mathcal{I}} + p)$, the budding conditional mean for exactly n released particles, with $d_{\mathcal{I}} + p = 8$ (the comparator scale of the source figure; any matched scale may be substituted and should then be stated in the caption). Optional second panel: three values of μ — $\lambda = 0.5$, $\delta = 0.01$, $\mu \in \{0, 0.005, 0.01\}$ — exact conditional means from simulation of the conditioned process, with the approximation $(\log n + \gamma - \mu/\lambda)/(\lambda + \delta - \mu)$ overlaid as a dashed curve for each. **Axes:** burst size n (horizontal), conditional mean rupture time (vertical); legend entries "exact", "harmonic approximation", "budding". **Style:** unified Python style — default matplotlib with `tab:` colours, grid alpha 0.25, `tight_layout`, PDF output, fonts matching the chapter. **Data source:** exact formula (43); approximation (44); budding comparator: $n+1$ exponential waiting times each of rate $d_{\mathcal{I}} + p$; optional panel by simulating the BDC process and conditioning on $\mathcal{K} = n$. **Placement:** Section 7.5, after Proposition 7.3.

8 Conditional burst means

The mean burst size and its various conditionings hide a small trap. It is worth springing it deliberately.

8.1 The overall expected burst size

Recall that $V(t) = \mathbb{E}(\mathbf{W}_t)$, where $\mathbf{W}_t$ is the number of particles released by time t. Whether the catastrophe has taken place or not can be represented by the truth of $\mathbf{W}_t > 0$, since the minimum burst size is 1; rupture has occurred with probability $1 - I(t)$. Decomposing,

$$\mathbb{E}(\mathbf{W}_t) = \mathbb{E}(\mathbf{W}_t \mid \mathbf{W}_t > 0)(1 - I(t)) + \mathbb{E}(\mathbf{W}_t \mid \mathbf{W}_t = 0)I(t),$$

and since $\mathbb{E}(\mathbf{W}_t \mid \mathbf{W}_t = 0) = 0$,

$$\mathbb{E}(\mathbf{W}_t) = \mathbb{E}(\mathbf{W}_t \mid \mathbf{W}_t > 0)(1 - I(t)),$$

that is,

$$\mathbb{E}(\mathbf{W}_t \mid \mathbf{W}_t > 0) = \frac{V(t)}{1 - I(t)}. \tag{49}$$

This can be read as "mean burst size, given that a burst has happened by t". The overall expected burst size — the expected size of an arbitrarily generated burst — is the late-time value of this expression:

$$\mathbb{E}(\mathcal{K}) = \lim_{t \to \infty} \frac{V(t)}{1 - I(t)} = \frac{V_\infty}{1 - b} = \frac{\lambda - \mu}{\delta} + \frac{1}{1 - b}, \tag{50}$$

where the last form follows from (12), and $\mathbb{E}(\mathcal{K}) = a/(a-1)$ as already noted there. The distinction matters: V_∞ itself averages in the realisations that cleared internally and released nothing, so V_∞ is *not* the mean burst size unless $\mu = 0$.

8.2 The curious circularity, resolved

What is the expected burst size of a cell that has *not yet* burst? Decomposing $\mathbb{E}(\mathcal{K})$ on the event $\{\tau > t\}$,

$$\begin{aligned}\mathbb{E}(\mathcal{K}) &= \mathbb{E}(\mathcal{K} \mid \tau > t)\Pr\{\tau > t\} + \mathbb{E}(\mathcal{K} \mid \tau < t)\Pr\{\tau < t\} \\ &= \mathbb{E}(\mathcal{K} \mid \tau > t)\, I(t) + \frac{V(t)}{1 - I(t)}\,(1 - I(t))\,, \end{aligned} \tag{51}$$

so that

$$\mathbb{E}(\mathcal{K} \mid \tau > t) = \frac{\mathbb{E}(\mathcal{K}) - V(t)}{I(t)} = \frac{V_\infty/(1-b) - V(t)}{I(t)}. \tag{52}$$

So far, so good. The trap is what happens when one takes $t \to \infty$ in (52). A first attempt gives

$$\lim_{t\to\infty} \mathbb{E}(\mathcal{K} \mid \tau > t) = \frac{V_\infty/(1-b) - V_\infty}{b},$$

which, after a little algebra, appears to return $\mathbb{E}(\mathcal{K})$ itself — a curious circularity: the burst size of a cell that has waited an infinite time to burst seems to equal the burst size of a typical cell, which makes no sense, since conditioning on $\tau > t$ as $t \to \infty$ should select the realisations that never burst at all.

The resolution is that (52) mixes two conditionings. Its numerator $\mathbb{E}(\mathcal{K}) - V(t)$ is the contribution to the burst size from realisations that burst *after* t; its denominator $I(t)$ is the probability of not having burst by t, which includes the mass b of realisations that will never burst and whose burst size is zero. Written consistently,

$$\mathbb{E}(\mathcal{K}\,\mathbf{1}\{\text{burst}\} \mid \tau > t) = \frac{V_\infty - V(t)}{I(t)} \xrightarrow[t\to\infty]{} \frac{V_\infty - V_\infty}{b} = 0, \tag{53}$$

which is correct and unmysterious: conditioning on "has not burst yet" converges to "never bursts", on which event the burst size is zero.

The genuinely interesting limit conditions on eventual burst as well:

$$\mathbb{E}(\mathcal{K} \mid \tau > t,\ \text{burst eventually}) = \frac{V_\infty - V(t)}{I(t) - b}. \tag{54}$$

This is of the form $0/0$ as $t \to \infty$; by l'Hôpital's rule, using $V' = \delta K$ and $I' = -\delta J$,

$$\lim_{t\to\infty} \mathbb{E}(\mathcal{K} \mid \tau > t,\ \text{burst eventually}) = \lim_{t\to\infty} \frac{K(t)}{J(t)} = \frac{\langle X^2\rangle_{\mathrm{QS}}}{\langle X\rangle_{\mathrm{QS}}} = 1 + \frac{2\lambda}{\delta}(1-b), \tag{55}$$

in agreement with (40): the eventual burst size of a very patient cell is the size-biased quasi-stationary mean. The circularity was never in the mathematics; it was in the conditioning.

9 Multiplicity of infection

So far the cell has been founded by a single particle. In applications this is not always the right picture: a macrophage may be invaded by several bacteria at once, and in the laboratory the multiplicity of infection is a control parameter. Nothing essential changes — the process is still linear — but the bookkeeping does, and one plausible guess turns out to be wrong. Throughout this section, k founders share a single cell and hence a single catastrophe clock: rupture occurs at rate δ times the *total* load, and dumps the whole cell.

9.1 Fixation functions for k founders

By the branching property, the k founder lineages evolve independently until the shared clock rings. No burst occurs by t precisely when no lineage has burst; internal extinction occurs precisely when every lineage has cleared internally. Hence

$$I_k(t) = I(t)^k, \qquad D_k(t) = D(t)^k, \qquad I_{\mathrm{fix},k}(t) = I(t)^k - D(t)^k. \tag{56}$$

Remark 9.1 (A guess that fails). The tempting guess is $I_{\mathrm{fix},k} = I_{\mathrm{fix}}^k$: each lineage stays productive with probability I_{fix}, independently. But "productive" for the whole cell means *not fixed*, and the intersection of the independent events "lineage i avoids bursting and does not go extinct" is not the event "the cell is not fixed": the cell also stops being productive when every lineage dies out. Correctly, $I_{\mathrm{fix},k} = I^k - D^k$, whereas $I_{\mathrm{fix}}^k = (I - D)^k$; already for $k = 2$ the two differ by $I^2 - D^2 - (I - D)^2 = 2D(I - D) \neq 0$ whenever $\mu > 0$. Numerically, at $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$ and $t = 1$: the correct $I_{\mathrm{fix},2} = I^2 - D^2 \approx 0.8458$, while $I_{\mathrm{fix}}^2 \approx 0.6542$. When $\mu = 0$, $D \equiv 0$ and the two coincide — again the invisible case.

9.2 Moments and the release kernel

The moments fall out of the k-founder probability generating function of §4.2, $G_k = G^k$, where G is the single-founder (defective) probability generating function. Differentiating at $z = 1$, with $G(1, t) = I(t)$, $\partial_z G|_{z=1} = J(t)$ and $\partial_z^2 G|_{z=1} = K(t) - J(t)$,

$$J_k(t) = k\, I(t)^{k-1} J(t), \tag{57}$$

and, from the second factorial moment,

$$\begin{aligned} K_k(t) &= \partial_z^2 G_k\big|_{z=1} + \partial_z G_k\big|_{z=1} \\ &= k(k-1) I(t)^{k-2} J(t)^2 + k I(t)^{k-1} \left(K(t) - J(t)\right) + k I(t)^{k-1} J(t) \\ &= k\, K(t) I(t)^{k-1} + k(k-1) J(t)^2 I(t)^{k-2}. \end{aligned} \tag{58}$$

The expected release flux — the quantity on which the population-level renewal model of Chapter 4b will be built — is therefore

$$g_k(t) = \delta K_k(t) = \delta \left(k\, K(t) I(t)^{k-1} + k(k-1) J(t)^2 I(t)^{k-2} \right). \tag{59}$$

Remark 9.2 (The free-sum trap). It is tempting to write $g_k = \delta\left(kK + k(k-1)J^2\right)$, the second moment of a free sum of k independent copies. That form omits the powers of I: the second-moment contribution of a configuration is counted only while the shared clock has *not* rung. At $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$, $t = 1$, $k = 2$: the free-sum form gives $K_2 \approx 23.39$, while (58) gives $K_2 \approx 22.264$, which agrees with a direct master-equation computation. The difference is the superlinearity of g_k in k wearing a disguise: even the free-sum form is superlinear, but the true kernel is superlinear *and* tempered by survival. That a cell's expected release flux depends superlinearly on its founding multiplicity is a genuine prediction of the model — one that classical constant-p models cannot express, and that low-multiplicity single-cell assays can test.

9.3 Lifetime yield from k founders

The lifetime mean release from k founders is $V_\infty^{(k)} = \int_0^\infty g_k(t)\,\mathrm{d}t$. Carrying out the integration (deferred to Appendix C),

$$V_\infty^{(1)} = V_\infty = \frac{a(1-b)}{a-1}, \tag{60}$$

and, for $k \geq 2$,

$$V_\infty^{(k)} = (1 - b^k) + \frac{2\lambda}{\delta}\left[\frac{1}{k+1} - b^k + \frac{k\, b^{k+1}}{k+1}\right] + k(k-1)\frac{\lambda}{\delta}\int_1^b (x-a)(x-b)\, x^{k-2}\, \mathrm{d}x. \tag{61}$$

In the no-death case the answer simplifies dramatically:

$$V_\infty^{(k)} = k + \frac{\lambda}{\delta} \qquad (\mu = 0). \tag{62}$$

Indeed, when $\mu = 0$ the event type is independent of the load (§10), so the number of births before the single catastrophe is $\mathrm{Geom}_0(s)$ with mean $\rho/s = \lambda/\delta$ whatever the founding load, and the total release is simply k founders plus that many births. For example, with $\lambda = 1$, $\delta = 0.1$: $V_\infty^{(k)} = 11, 12, 13, 14, 15, \ldots$

Two biological points are worth pausing on. First, the yield is *not* kV_∞: the first catastrophe dumps the entire cell once, so additional founders add their own expected descendants but not another cell's worth of release. At $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$: $V_\infty \approx 13.99$, so a naive scaling would predict $2V_\infty \approx 27.97$ for two founders; the true value is $V_\infty^{(2)} \approx 17.43$. Second, while the unconditional yield grows sub-extensively in k, the *conditional* mean burst size given eventual burst, $V_\infty^{(k)}/(1 - b^k)$, *increases* with k (17.23, 18.07, 19.02, 21.00 for $k = 1, 2, 3, 5$ at the same parameters): multiply infected cells are more likely to burst at all, and when they do, they burst bigger.

FIGURE FLAG F4a.6 — Multiplicity of infection

Purpose: display the two MOI effects of this section — the superlinear release flux in founding multiplicity, and the increasing conditional mean burst — in one figure. **Type:** Python analytic plot, two panels (or inset). **Content:** panel (a): the release fluxes $g_k(t) = \delta K_k(t)$ against age $t \in [0, 15]$ for $k = 1, 2, 3, 4$, with $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$, where $K_k = kKI^{k-1} + k(k-1)J^2 I^{k-2}$ from (58) and I, J, K are the closed forms of Chapter 3 (Eq. (7)–(9)); annotate the superlinear ordering at each t. Panel (b): the conditional mean burst $V_\infty^{(k)}/(1 - b^k)$ against $k = 1, \ldots, 6$ from (61), with the horizontal dashed line at $V_\infty/(1 - b) = a/(a - 1)$ (the $k = 1$ value) to show the increase. **Axes:** panel (a): age a (horizontal; label the axis "age" or "time since infection" to avoid collision with the root a — use "age" on the axis), release flux g_k (vertical); panel (b): founders k (horizontal), conditional mean burst (vertical); legend entries $k = 1, 2, 3, 4$. **Style:** unified Python style — default matplotlib with `tab:` colours, grid alpha 0.25, `tight_layout`, PDF output, fonts matching the chapter. **Data source:** analytic formulas (58), (61); no simulation. **Placement:** Section 9, at the end of the section.

10 Chained immediate transfer

10.1 Setup

We now couple several cells together in the most austere way possible. Take M host cells, all empty save one, which holds a single particle. Each cell runs the $\mu = 0$ birth–catastrophe process of the previous sections (birth λn, catastrophe δn); on rupture, however, the released load does not enter an extracellular pool. It is transferred immediately and wholesale into a fresh cell, where a new BDC process begins; and so on, until every cell has ruptured. There is no extracellular phase, no clearance, and no re-infection — only a chain of bursts.

The motivation is twofold. First, founding censuses can be remarkably small: in plague, say, colony-forming units are counted in the hundreds against a vast population of macrophages, so a serial, one-cell-at-a-time picture is not absurd as a caricature of early pathogenesis. Second, the model is a tractable laboratory for multi-cell burst statistics — the sizes and times of successive ruptures — which interpolate between the single-cell theory above and the population models of Part II. Since $\mu = 0$, internal extinction is impossible and all M ruptures occur almost surely.

Let r_i be the load at the ith rupture (the ith *rupture size*), t_i its time, and $T_i = t_i - t_{i-1}$ the ith inter-rupture interval (with $t_0 = 0$). Write

$$s = \frac{\delta}{\lambda + \delta}, \qquad \rho = \frac{\lambda}{\lambda + \delta} = 1 - s. \tag{63}$$

10.2 The key observation: event types ignore the load

When the load is n, the next event is a birth with probability $\lambda n/(\lambda n + \delta n) = \rho$ and a catastrophe with probability s — independent of n. Only the holding time depends on the load: it is $\operatorname{Exp}((\lambda + \delta)n)$. Consequently the sequence of event *types* is a sequence of independent coin flips, and the number of births G that occur before the catastrophe in a cell founded with any load is

$$G \sim \operatorname{Geom}_0(s), \qquad \Pr\{G = g\} = s\,\rho^g, \qquad g = 0, 1, 2, \ldots \tag{64}$$

(the number of failures before the first success). Each cell in the chain contributes an *independent* copy G_i of this variable — the Markov process restarts afresh at every transfer — and the rupture size of cell i is its founding load plus its own births. Since the first cell is founded by a single particle,

$$r_k = 1 + G_1 + \cdots + G_k, \tag{65}$$

with $G_1, \ldots, G_k$ i.i.d. $\operatorname{Geom}_0(s)$. Everything about the rupture sizes follows from this decomposition.

10.3 Rupture-size laws

The sum of k i.i.d. geometric variables is negative binomial, so (65) gives immediately:

$$\Pr\{r_k = n\} = \mathcal{A}_k^{(n)}\, s^k\, \rho^{\,n-1}, \qquad \mathcal{A}_k^{(n)} = \binom{n+k-2}{k-1}, \quad n \geq 1. \tag{66}$$

The first few cases display the pattern:

$$\Pr\{r_1 = n\} = s\rho^{n-1} \text{ (geometric — matches one-cell BDC)}, \tag{67}$$

$$\Pr\{r_2 = n\} = n\, s^2 \rho^{n-1}, \qquad \Pr\{r_3 = n\} = \tfrac{n(n+1)}{2} s^3 \rho^{n-1}, \qquad \Pr\{r_4 = n\} = \tfrac{n(n+1)(n+2)}{6} s^4 \rho^{n-1}. \tag{68}$$

Equivalently, the coefficients satisfy the recurrence

$$\mathcal{A}_k^{(n)} = \sum_{i=1}^{n} \mathcal{A}_{k-1}^{(i)}, \qquad \mathcal{A}_1^{(i)} = 1, \tag{69}$$

which is just (66) conditioned on the value of r_{k-1}: $\Pr\{r_k = n\} = \sum_i \Pr\{r_{k-1} = i\} \Pr\{G_k = n - i\}$. The closed form $\mathcal{A}_k^{(n)} = \binom{n+k-2}{k-1}$ then follows by induction from the hockey-stick identity — supplying the proof that the original investigation had conjectured but not completed.

Remark 10.1 (The coefficient, and a near miss). The coefficient in (66) is easy to miscount. The superficially similar expression $\frac{1}{k!}\frac{(n+k-1)!}{(n-1)!} = \binom{n+k-1}{k}$ is off by one shift, and is already contradicted by the r_2 and r_3 cases displayed above, which (66) reproduces exactly. The correct coefficient is $\binom{n+k-2}{k-1}$, and it has a direct reading: it counts the ways of distributing the $n-1$ births accumulated along the chain among its k cells, each cell receiving any non-negative number — the stars-and-bars count for (65).

Further consequences, all verified numerically (see §10.7):

$$\mathbb{E}(r_k) = 1 + k\,\frac{\rho}{s} = 1 + k\,\frac{\lambda}{\delta}, \qquad \mathrm{Var}(r_k) = k\,\frac{\rho}{s^2}, \tag{70}$$

$$\mathbb{E}(z^{r_k}) = z\Big(\frac{s}{1-\rho z}\Big)^k, \qquad \sum_{n\geq 1} \Pr\{r_k = n\} = 1 \text{ (burst is certain for } \mu = 0). \tag{71}$$

The mean rupture size grows linearly along the chain — each cell adds one expected burst's worth of births, λ/δ, to whatever it inherited.

10.4 Rupture times

The first rupture time inherits the single-cell flux. With m founders, the no-rupture probability is $I(t)^m$ by (56) (with $\mu = 0$, $D \equiv 0$), so

$$\Pr\{t_1 > t\} = I(t)^m, \qquad f_{t_1}^{(m)}(t) = -m\,I(t)^{m-1} I'(t) = m\,I(t)^{m-1}\delta J(t). \tag{72}$$

For $m = 1$ this is the burst-time density $\varphi = \delta J$ of §7.1 (which is proper here, since $b = 0$). When $\mu > 0$, the same reasoning distinguishes the two fixation routes, as in the original investigation: the *burst-only* time has CDF $(1 - I(t))/(1 - b)$ and density $\delta J(t)/(1 - b)$, while the *general-fixation* time — rupture or internal extinction — has density

$$\delta J(t) + \mu\, p_1(t), \tag{73}$$

the extra term being the flux through $1 \to 0$: only a population of size exactly one can die out in a single death event.

10.5 Inter-rupture intervals

Given k founders, the interval $T(k)$ until rupture decomposes over the holding times: if $G = g$ births precede the catastrophe, the load runs through $k, k+1, \ldots, k+g$, so

$$T(k) = \sum_{j=k}^{k+G} \xi_j, \qquad \xi_j \sim \mathrm{Exp}\left((\lambda+\delta)j\right) \text{ independent.} \tag{74}$$

Averaging over $G \sim \mathrm{Geom}_0(s)$, the Laplace transform is

$$\widetilde{T}(k,u) = \mathbb{E}(e^{-uT(k)}) = \sum_{g\geq 0} s\rho^g \prod_{j=k}^{k+g} \frac{(\lambda+\delta)j}{(\lambda+\delta)j + u}. \tag{75}$$

Writing $u' = u/(\lambda+\delta)$ and unrolling the product with Gamma functions, the series sums to a hypergeometric form — a theme that recurs in Chapter 4b, where the effective parameters of the population model are likewise expressed through hypergeometric Laplace transforms:

$$\widetilde{T}(k,u) = \frac{s\,k}{k+u'}\,{}_2F_1\Big(k+1, 1; k+1+u'; \rho\Big). \tag{76}$$

Differentiating (75) at $u = 0$ gives the exact mean

$$\mathbb{E}(T(k)) = \sum_{m\geq 0} \frac{\rho^m}{(\lambda+\delta)(k+m)}, \tag{77}$$

which is strictly decreasing in k: the more a cell inherits, the faster it bursts. This quantifies the original investigation's qualitative observation that the chain is "partitioned by reducing intervals of time". For $(\lambda, \delta) = (1, 1)$: $\mathbb{E}(T(1..4)) \approx 0.693, 0.386, 0.273, 0.212$ for fixed founding loads; the chain intervals T_k, whose founding loads are random, decrease faster still $(0.691, 0.500, 0.377, 0.292$ in simulation).

10.6 The second rupture time

The chain's dependence structure is now visible. The second rupture time is $t_2 = t_1 + T_2$, but t_1 and T_2 are *not* independent: both are functions of G_1, the number of births in the first cell. Conditional on $G_1 = g$, however, the two pieces involve disjoint sets of exponential holding times, so their Laplace transforms factor:

$$\mathbb{E}(e^{-ut_2} \mid G_1 = g) = P_1(1+g, u)\, \widetilde{T}(1+g, u), \qquad P_1(m, u) = \prod_{j=1}^{m} \frac{(\lambda+\delta)j}{(\lambda+\delta)j + u}. \tag{78}$$

Averaging over $G_1 \sim \mathrm{Geom}_0(s)$:

$$\widetilde{t_2}(u) = \sum_{g \geq 0} s\rho^g\, P_1(1+g, u)\, \widetilde{T}(1+g, u). \tag{79}$$

This resolves the question with which the original investigation closed. The convolution guessed there, $\rho(t_2) = \int f(\tau) \sum_k \Pr\{r_1 = k|\tau\} \phi_k(t_2 - \tau) \mathrm{d}\tau$, is the same statement in different clothes — and, as suspected, it is not the tractable one. The Laplace form (79) is; so is the density itself, computable exactly as a mixture of hypoexponential densities ($t_1|G_1$ and $T_2|(G_1, G_2)$ are sums of independent exponentials) and verified against simulation in §10.7. Moments assemble from the pieces, e.g. $\mathbb{E}(t_2) = \mathbb{E}(T(1)) + \sum_{n \geq 1} \Pr\{r_1 = n\} \mathbb{E}(T(n))$.

10.7 Verification record and scope

Every quantitative claim of this section was checked by exact simulation of the chained process (the event-type decomposition makes sampling direct): the suite `verify_chained_transfer.py` comprises 28 checks across three parameter sets — the coefficient identity (exact integer arithmetic); the rupture-size laws for $k = 1..4$ against 10^6 chains; moments (70); the probability generating function (71); the t_1 density (72) vs δJ; the interval transform (75) for fixed founders $k \in \{1, 2, 3, 5\}$; the random-founder mixture consistency of the chain intervals; the t_2 transform (79); the exact hypoexponential-mixture density of t_2 (relative $L^2 \approx 3\%$ against histograms); and the monotone interval means. **28 checks, all passing.**

What remains open, and is stated plainly: the joint laws of (t_k, r_k) for general $\mu > 0$ (a coefficient machinery with ${}_2F_1$ antiderivatives is the route, and the single-cell ingredients are in Appendix C), and the joint law of (t_M, r_M) for the full M-cell chain. For $\mu = 0$, the decomposition (65) extends the size laws to all M at once, since nothing about it depends on where a cell sits in the chain.

FIGURE FLAG F4a.7 — Chained transfer

Purpose: show the section's two headline structures — the negative-binomial rupture-size laws and the decreasing inter-rupture intervals — in one figure. **Type:** Python simulation + analytic overlay, two panels. **Content:** panel (a): rupture-size pmfs $\Pr\{r_k = n\}$ for $k = 1, 2, 3, 4$ with $(\lambda, \delta) = (1, 1)$ (so $s = \rho = 1/2$): histogram bars from exact simulation of 10^6 chains (the event-type decomposition makes sampling direct), with the closed forms $\binom{n+k-2}{k-1} s^k \rho^{n-1}$ from (66) overlaid as connected curves. Panel (b): mean inter-rupture intervals $\mathbb{E}(T(k)) = \sum_{m \geq 0} \rho^m / ((\lambda + \delta)(k + m))$ from (77) against $k = 1, \ldots, 6$, plotted as a decreasing curve, with simulated chain-interval means (random founders) overlaid as points to show they decrease faster. Adapt the exact-sampling machinery of the verification suite (do not modify the suite itself; write a chapter-local script). **Axes:** panel (a): rupture size n (horizontal), probability (vertical), legend $k = 1..4$; panel (b): k (horizontal), mean interval (vertical). **Style:** unified Python style — default matplotlib with `tab:` colours, grid alpha 0.25, `tight_layout`, PDF output, fonts matching the chapter. **Data source:** closed forms (66), (77); simulation via the event-type decomposition (G_i i.i.d. $\mathrm{Geom}_0(s)$; exponential holding times of rate $(\lambda + \delta)n$).
Placement: Section 10, after the verification record.

11 Discussion

This chapter set out to complete the single-cell theory of the birth–death–catastrophe process, and that programme is now done: the single-cell description is complete in distribution — geometric states at every time, a geometric quasi-stationary limit, and a burst-size law identical to it — with the conditional laws of burst time and burst size, the multi-founder extension, and the chained-transfer model alongside. What remains is to say what the results imply for measurement, where they have already been felt in real systems, and what is still open on the single-cell side. The population consequences — fitting, thresholds, and the comparison of bursting with budding — are taken up in Chapter 4b.

11.1 Single-cell assay predictions for lytic systems

For a genuinely lytic system — *Y. pestis*, *F. tularensis*, lytic phage — a limiting-dilution assay (one infected cell per well, counts followed in time) measures the single-cell process directly, with no population model interposed. The number of new infectious particles released per infected cell is an integer-valued random variable, and its full distribution, not merely its mean, is what determines whether a small inoculum grows into a detectable infection [6]; assays of this design are where that distribution becomes observable. The theory makes five predictions, all of them parameter-free given (λ, μ, δ):

1. a fraction b of wells show no release ever — the mass of internal extinction;
2. the cumulative release in a well is a step function: zero, then a single terminal jump;
3. the jump times follow the density $\delta J(t)/(1 - b)$;
4. the jump sizes follow the geometric law with ratio $1/a$ — the quasi-stationary distribution, by Corollary 7.2;
5. the between-well variance of total release is $\mathrm{Var}(W_\infty)$ of (13), a quantity no deterministic constant-rate model can produce at all.

One honest caveat: these predictions are too strong for a budding virus as literally stated — intermittent budding is not a single terminal burst — and they should be used for lytic hosts, or as the bursting endpoint of the spectrum of release models that Chapter 4b lays out.

11.2 The same theory at two extremes

The single-cell theory of this chapter has already been fitted to two intracellular pathogens whose parameter estimates sit at opposite ends of what the model allows, and the contrast is worth recording here because it shows how much of the behaviour is fixed by (λ, μ, δ) alone.

For *F. tularensis* strain SCHU S4, modelling of murine challenge data yielded the estimates $\lambda = 0.15\,\mathrm{h}^{-1}$, $\mu = 0.01\,\mathrm{h}^{-1}$ and $\delta = 1.5 \times 10^{-4}\,\mathrm{h}^{-1}$ [7]. In the model with these parameters, fewer than 7% of infected cells fail to burst — the internal-extinction probability b is small — and the burst sizes of the cells that do rupture are geometrically distributed with typical sizes in the hundreds of bacteria. That prediction sits comfortably with the observed ability of doses as low as 10 colony-forming units to establish infection [8]: each successful rupture floods the local environment with enough bacteria that a small inoculum needs only a few rounds of the cycle to reach large numbers. Recent video recordings of *F. tularensis*-infected macrophages, showing en-masse release events and direct counts of near-rupture payloads, provide observational support for the burst-size prediction.

For *Bacillus anthracis*, approximate Bayesian computation against published experimental time-series produced the very different estimates $\lambda \approx 0.64\,\mathrm{h}^{-1}$, $\mu \approx 1.64\,\mathrm{h}^{-1}$ and $\delta \approx 0.04\,\mathrm{h}^{-1}$ [9]. Here death dominates birth: the model predicts that more than 96% of phagocytes clear their intracellular infection without rupturing, and the cells that do rupture release on average only about 1.6 bacteria each — the value of $\mathbb{E}(\mathcal{K} \mid \text{burst}) = a/(a-1)$ at these parameters. The consistent picture is that an infectious dose of 8×10^3 to 5×10^4 spores is required [9], orders of magnitude above the tularemia figure, and one consistent with a pathogen whose intracellular dynamics destroy far more than they release.

These are model-based predictions, and the biological claims rest on the parameter estimates they condition on; but the two cases together show that the same three-rate theory spans the space from near-certain, large bursts to near-certain clearance with minimal release, and that the quantities fitted in each case — the no-burst fraction b, the burst-size law, the mean conditional burst — are exactly the assay observables listed above.

11.3 Open problems

Three questions remain open on the single-cell side.

1. **A conceptual proof that burst size equals QSD.** The algebra of §7.3 is complete; a structural argument — presumably size-biasing composed with the burst intensity, or a Yaglom-type limit [5] — would turn a coincidence into a theorem one can quote.

2. **Chained joint laws for general μ.** The $\mu = 0$ chain of §10 rests on the event-type decomposition, which fails once internal extinction is possible. The joint laws of (t_k, r_k) for $\mu > 0$ — and the joint law of (t_M, r_M) for the full chain — remain open; a coefficient machinery with ${}_2F_1$ antiderivatives is the route sketched in §10.

3. **Logistic intracellular growth.** A referee will object that linear birth lets the conditional load grow without bound. The defence is that the burst hazard is proportional to load and so regulates it — the QSD result is precisely the statement that the conditional load stabilises at $a/(a-1)$ — but the sensitivity of the burst statistics, and of the kernels that Chapter 4b will use, to a logistic birth rate remains to be checked numerically.

What comes next

Everything in this chapter is single-cell. The released particles have so far been counted and then forgotten, and that is the approximation the next chapter removes. In Chapter 4b, the survival and release statistics derived here become the age-structured kernels of a population-level renewal model: cells infected at different times contribute to the infected-cell and free-particle pools according to $I_{\text{fix}}(a)$ and $g(a) = \delta K(a)$, the constant-rate models of viral dynamics appear as a projection of that renewal system, and the bursting strategy is compared against budding at the level where the comparison matters — establishment, invasion speed, and what data can and cannot identify.

A Master formula table: single cell

Throughout: $w = \mathrm{e}^{\theta t}$, $\theta = \lambda(a-b)$, $A = a-1$, $B = 1-b$, $\kappa = 1 + \delta/(2\lambda)$.

Quantity	Closed form
Roots a, b	$\eta \pm \sqrt{\eta^2 - \mu/\lambda}$, $\eta = (\lambda+\mu+\delta)/(2\lambda)$, $b < 1 < a$
A, B, θ	$a-1$, $1-b$, $\lambda(a-b)$; $AB = \delta/\lambda$, $A+B = a-b$
$I(t)$	$(aB + bAw)/(B + Aw)$
$D(t)$	$ab(w-1)/(aw-b)$
$I_{\text{fix}}(t)$	$(a-b)^2 w/[(B+Aw)(aw-b)]$
$I_{\text{fix},k}(t)$	$I(t)^k - D(t)^k$
$J(t)$	$(a-b)^2 w/(B+Aw)^2 = -\frac{\lambda}{\delta}(I-a)(I-b)$
$K(t)$	$[1 + 2\lambda(1-I)/\delta]J = \frac{2\lambda}{\delta}(\kappa - I)J$
$V(t)$	$(1-I)[1 + \lambda(1-I)/\delta]$
V_∞	$a(1-b)/(a-1) = aB/A$
$\mathbb{E}(\mathcal{K} \mid \text{burst})$	$a/(a-1)$
$\mathbb{E}({\mathbf{W}_t}^2)$	$\frac{2(\lambda-\mu)}{\delta}V - \frac{\lambda+\mu}{\delta}I - K + \frac{\lambda+\mu}{\delta} + 1$
$\varphi(t)$ (burst time)	$\delta J(t)$, mass $1-b$
$\mathbb{P}\{\mathcal{K} = k\}$	$(\delta/\lambda)\, a^{-k}$, $k \geq 1$
$\mathbb{P}\{\mathcal{K} = k \mid \text{burst}\}$	$(a-1)\, a^{-k}$, $k \geq 1$
$\mathrm{QSD}(n)$	$(a-1)\, a^{-n}$
$\mathbb{E}(T_{\text{prod}})$	$\lambda^{-1}\log(a/(a-1)) = \lambda^{-1}\log\langle X\rangle_{\text{QS}}$
$\mathbb{E}(\tau \mid \mathcal{K} = n)$ $(\mu = 0)$	$\theta^{-1}\sum_{k=1}^{n} 1/k$, $\theta = \lambda + \delta$
$\mathbb{E}(\tau \mid \text{burst})$ $(\mu > 0)$	$\frac{1}{\lambda(1-b)}\log\frac{a-b}{a-1}$
$J_k(t)$ (MOI)	kJI^{k-1}
$K_k(t)$ (MOI)	$kKI^{k-1} + k(k-1)J^2 I^{k-2}$
$g_k(t)$ (MOI)	$\delta K_k(t)$
$V_\infty^{(k)}$, $\mu = 0$	$k + \lambda/\delta$
r_k law ($\mu = 0$, chain)	$\binom{n+k-2}{k-1} s^k \rho^{n-1}$, $s = \delta/(\lambda+\delta)$
$\mathbb{E}(r_k)$, $\mathrm{Var}(r_k)$	$1 + k\lambda/\delta$, $k\rho/s^2$
$\widetilde{T}(k, u)$ (interval)	$\frac{sk}{k+u'}\,{}_2F_1(k+1, 1; k+1+u'; \rho)$, $u' = u/(\lambda+\delta)$

The $\mu = 0$ specialisation. $b = 0$, $a = 1 + \delta/\lambda$, $D \equiv 0$, $I_{\text{fix}} = I$, $V_\infty = 1 + \lambda/\delta = \langle X \rangle_{\text{QS}}$, $\mathbb{E}(T_{\text{prod}}) = \lambda^{-1} \log(1 + \lambda/\delta)$, $V_\infty^{(k)} = k + \lambda/\delta$, and $\mathbb{P}\{\mathcal{K} = k\} = s\rho^{k-1}$ with $s = \delta/(\lambda + \delta)$, $\rho = 1 - s$: the burst size is geometric on $\{1, 2, \ldots\}$ with success probability s, and every realisation bursts.

B Verification record: chained immediate transfer

Every quantitative claim of §10 was checked by exact simulation of the chained process — the event-type decomposition of §10.2 makes sampling direct. This appendix records the suite, its tally, and how to reproduce it.

The suite `verify_chained_transfer.py` samples the chained process exactly and checks every quantitative claim across three parameter sets, $(\lambda, \delta) \in \{(1, 1), (1, 0.1), (2, 0.5)\}$: 28 checks, all passing.

ID	Name	What is checked	Result
A	Coefficient identity	$\binom{n+k-2}{k-1}$ obeys the recurrence (exact integers)	1/1
B	Rupture-size laws	$\mathbb{P}\{r_k = n\}$ vs 10^6 chains, $k = 1..4$, SE-unit comparison	3/3
C	Moments	$\mathbb{E}(r_k) = 1 + k\lambda/\delta$, $\mathrm{Var}(r_k) = k\rho/s^2$	3/3
D	PGF	$\mathbb{E}(z^{r_k}) = z(s/(1 - \rho z))^k$ (SE units)	3/3
E	First rupture time	histogram vs $\delta J(t)$	3/3
F	Interval transform	fixed founders $k \in \{1, 2, 3, 5\}$: empirical Laplace transform vs the series / ${}_2F_1$ form	3/3
F2	Interval mixture	chain intervals T_2, T_3 (random founders) vs pmf-weighted $\widetilde{T}$	3/3
G	Second rupture time	empirical $\mathbb{E}(e^{-ut_2})$ vs $\widetilde{t_2}(u)$	3/3
H/H2	t_2 density and mean	exact hypoexponential-mixture density vs histogram (rel. $L^2 \approx 3\%$); exact $\mathbb{E}(t_2)$	3/3
I	Interval means	strictly decreasing; simulation vs the exact series	3/3
	Total		**28/28**

The ${}_2F_1$ cross-check in test F activates only when scipy is available; the series itself (which the simulation is compared against) needs only numpy.

Reproduction.

```
cd "4 BDC additional and BMVR/N=2 Immediate Transfer content"
python3 verify_chained_transfer.py
# writes verify_chained_transfer_report.txt, metrics, figures
```

C Technical derivations

C.1 Lifetime yield from k founders

We derive (61). Start from $V_\infty^{(k)} = \int_0^\infty g_k(t)\,\mathrm{d}t$ with g_k from (59), and change variables from t to $x = I(t)$: as t runs $0 \to \infty$, I runs $1 \to b$, and since $\mathrm{d}I/\mathrm{d}t = -\delta J$, $\mathrm{d}t = -\mathrm{d}x/(\delta J)$. The first part of

g_k gives

$$\int_0^\infty \delta k K I^{k-1}\mathrm{d}t = \int_b^1 k\,\frac{K}{J}\,x^{k-1}\mathrm{d}x = \int_b^1 k\Big(1+\frac{2\lambda}{\delta}-\frac{2\lambda}{\delta}x\Big)x^{k-1}\mathrm{d}x,$$

using $K/J = 1+2\lambda(1-I)/\delta$. Integrating,

$$= \Big(1+\frac{2\lambda}{\delta}\Big)(1-b^k) - \frac{2\lambda k}{\delta}\,\frac{1-b^{k+1}}{k+1} = (1-b^k) + \frac{2\lambda}{\delta}\Big[\frac{1}{k+1} - b^k + \frac{k\,b^{k+1}}{k+1}\Big],$$

where the second line collects terms over the common denominator $k+1$. The second part of g_k gives, with $J = -\frac{\lambda}{\delta}(I-a)(I-b)$,

$$\int_0^\infty \delta k(k-1)J^2 I^{k-2}\mathrm{d}t = \int_b^1 k(k-1)J\,x^{k-2}\mathrm{d}x = k(k-1)\frac{\lambda}{\delta}\int_1^b (x-a)(x-b)x^{k-2}\mathrm{d}x,$$

on reversing the limits. Adding the two parts yields (61). The formula also holds at $k=1$ (the integral term vanishes), reducing to $V_\infty = (1-b)[1+\lambda(1-b)/\delta]$ of (12). When $\mu=0$, $b=0$ and the integral simplifies to $k(k-1)\frac{\lambda}{\delta}\int_1^0 (x-a)x^{k-1}\mathrm{d}x = k(k-1)\frac{\lambda}{\delta}\big[\frac{a-1}{k} - \frac{1}{k+1}\big]$, which combines with the first part to give $k+\lambda/\delta$ — the elementary result of §9, recovered analytically.

C.2 PGF coefficient extraction for general initial conditions

The general-μ route for the chained model of §10 runs through the k-founder probability generating function. From §4.2 (catastrophe convention),

$$G_k(z,t) = \left[\frac{ab\gamma(t)+v(t)z}{1-\gamma(t)z}\right]^k, \qquad \gamma(t) = \frac{1-\mathrm{e}^{\theta t}}{b-a\mathrm{e}^{\theta t}}, \qquad v(t) = \frac{b\mathrm{e}^{\theta t}-a}{b-a\mathrm{e}^{\theta t}}.$$

Writing $\Theta_m = \binom{k}{m}(ab)^{k-m}\gamma(t)^{k-m}v(t)^m$ and $\Omega_n = \binom{k-1+n}{n}\gamma(t)^n$, the binomial expansion of the numerator and the negative-binomial series of $(1-\gamma z)^{-k}$ give $G_k(z,t) = (\sum_{m=0}^k \Theta_m z^m)(\sum_{n\ge 0}\Omega_n z^n)$, so the state-probability coefficients are, with H the power of z,

$$[z^H]G_k = \sum_{i=0}^{\min(H,k)} \Theta_{k-i}\,\Omega_{i+\max(0,H-k)}.$$

The chained burst-size laws then require integrals of the form $\int_0^\infty \gamma(t)^n v(t)^m \mathrm{d}t$ for arbitrary $n,m\in\mathbb{N}$. Binomial-expanding the two numerators,

$$(1-\mathrm{e}^{\theta t})^n(b\mathrm{e}^{\theta t}-a)^m = \sum_{i=0}^{n+m} K_i\,(\mathrm{e}^{\theta t})^i$$

for explicit constants K_i, the problem reduces to the elementary building blocks

$$\int \frac{(\mathrm{e}^{\theta t})^i}{(b-a\mathrm{e}^{\theta t})^j}\mathrm{d}t = -\frac{i\,(\mathrm{e}^{\theta t})^i\,(b-a\mathrm{e}^{\theta t})^{1-j}}{b\,\theta}\,{}_2F_1\Big(1,(1+i)-j;1+i;\frac{a\mathrm{e}^{\theta t}}{b}\Big),$$

and the burst-size law for the kth rupture follows by summing over the initial-condition distribution. For $\mu=0$ this entire machinery collapses to the geometric decomposition of §10.2; for $\mu>0$ it remains the available route, and the joint laws of (t_k, r_k) are the open problem stated in §10.7.

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